GRAVITYENGINE · BSDT · PHASE GEOMETRY

Mathematical Anatomy of System Collapse

Closed-form analytical formulas that replace running the simulation — direct computation of every output from data alone

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§ I

State Space & Notation

The system consists of N agents evolving in $\mathbb{R}^d$. At each discrete time t the full configuration is a matrix:

STATE MATRIX

$$X_t \in \mathbb{R}^{N \times d}, \qquad X_t = \bigl[x_t^{(1)}, \ldots, x_t^{(N)}\bigr]^\top$$

N	Number of agents (institutions)
d	State-space dimension per agent
$\mu \in \mathbb{R}^d$	Equilibrium / target centroid
T	Trajectory length (time steps)
x_i	Row *i* of X — state of agent i
r_{ij}	$\|x_i - x_j\|$, Euclidean distance between agents

§ II

Inter-Institution Correlation Network

Before forces are computed, the coupling topology of the system is established via a Ledoit-Wolf shrinkage correlation network built on the leverage feature $\ell_t^{(i)} = X_t^{(i,0)}$ (loan-to-asset ratio, feature index 0).

2.1 — Sample Covariance

CENTRED SAMPLE COVARIANCE

$$S = \frac{1}{T}\sum_{t=1}^{T}(\ell_t - \bar\ell)(\ell_t - \bar\ell)^\top \in \mathbb{R}^{N\times N}, \qquad \bar\ell = \frac{1}{T}\sum_t \ell_t$$

2.2 — Ledoit-Wolf Analytical Shrinkage

The shrinkage target is the scaled identity — the minimum-assumption prior that all institutions are equally variant and uncorrelated:

SHRINKAGE TARGET (SCALED IDENTITY)

$$T^* = \frac{\operatorname{tr}(S)}{N}\,I_N$$

ORACLE APPROXIMATING SHRINKAGE INTENSITY

$$\rho = \operatorname{clip}\!\left(\frac{\rho_{\text{num}}}{\rho_{\text{den}}},\; 0,\; 1\right), \qquad \rho_{\text{num}} = \frac{1}{T^2}\sum_{i,j}\operatorname{Var}(X_{ti}X_{tj}), \qquad \rho_{\text{den}} = \|S - T^*\|_F^2$$

Implemented as:

$$\rho_{\text{num}} = \frac{1}{T}\left(\frac{X_c^{\odot 2\top} X_c^{\odot 2}}{T} - S^{\odot 2}\right)_{\!\text{sum}}\!\Big/\,T, \quad X_c^{\odot 2} = X_c \odot X_c \text{ (elementwise square)}$$

USE INSTEAD OF SIMULATION

CLOSED-FORM P — DIRECT FROM LEVERAGE PANEL, NO ITERATION

$$\rho = \operatorname{clip}\!\left(\frac{\displaystyle\frac{1}{T^2 N^2}\,\bigl\|X_c^{\odot 2}\bigr\|_F^2 - \frac{1}{T N^2}\|S\|_F^2} {\displaystyle\|S\|_F^2 - \frac{[\operatorname{tr}(S)]^2}{N}},\; 0,\;1\right)$$

Scalar interpretation: \(\rho \approx

0) when $T \gg N$ (data-rich regime — trust empirical covariance); $\rho \to 1$ when $T \ll N$ (data-starved — pull toward independence prior). No optimisation loop required: compute $\|X_c^{\odot 2}\|_F^2$, $\|S\|_F^2$, and $\operatorname{tr}(S)$ directly from the panel.

Inputs needed: Centred leverage panel $X_c \in \mathbb{R}^{T\times N}$, sample covariance S

SHRUNK COVARIANCE

$$\hat\Sigma = (1-\rho)\,S + \rho\,T^*$$

2.3 — Correlation Matrix W

The network adjacency weights are partial correlations, symmetrised and unit-diagonal:

NORMALISATION TO CORRELATION

$$W_{ij} = \frac{\hat\Sigma_{ij}}{\sqrt{\hat\Sigma_{ii}\,\hat\Sigma_{jj}}}, \qquad W_{ii} = 1, \qquad W = \frac{W + W^\top}{2}$$

2.4 — Network Spectral Radius

SPECTRAL RADIUS OF THE NETWORK

$$\rho(W) = \lambda_{\max}(W) = \max_k |\lambda_k|, \qquad W v_k = \lambda_k v_k$$

Computed via symmetric eigensolver (`eigh`).

For symmetric positive semi-definite W, all eigenvalues are real and $\lambda_{\max}(W) = \rho(W)$. As correlations increase, $\lambda_{\max}$ approaches N — perfect synchrony — the maximum contagion amplification factor.

USE INSTEAD OF SIMULATION

CLOSED-FORM BOUNDS — NO EIGENSOLVER NEEDED

$$\bar{W}_{\text{off}} \leq \frac{\rho(W) - 1}{N-1} \leq W_{\text{max}}$$ $$\rho(W) \leq \sqrt{1 + (N-1)\,W_{\text{max}}^2\,(N-1)} \quad\text{(Frobenius bound)}$$ **Quick scalar proxy** (replaces full eigen-decomposition for early-warning threshold checks): $$\tilde\rho = 1 + (N-1)\,\bar{W}_{\text{off}}$$ where $\bar{W}_{\text{off}} = \frac{1}{N(N-1)}\sum_{i\neq j}W_{ij}$ is the mean off-diagonal correlation. When $\tilde\rho >

N/2\), the threshold for network-amplified contagion is breached without computing a single eigenvalue.

`Inputs needed:` Mean off-diagonal $\bar W_{\text{off}}$, max off-diagonal W_{max}, N

`ρ(W) = 1`	Uncorrelated institutions — no network amplification
`ρ(W) → N`	Perfect synchrony — maximum systemic contagion
`ρ`	Ledoit-Wolf shrinkage intensity ∈ [0,1]; higher ρ pulls toward independence prior
`W̄_off`	Mean off-diagonal correlation — average bilateral coupling
`W_max`	Maximum off-diagonal correlation — strongest bilateral link

2.5 — Relationship to GravityEngine Coupling

The network spectral radius $\rho(W)$ is the macro-level diagnostic; the pairwise energy coupling constant γ in Φ_{pair} is the micro-level parameter. They connect conceptually as:

COUPLING CORRESPONDENCE

$$\gamma \;\longleftrightarrow\; \frac{\rho(W)}{N} \cdot \gamma_{\text{scale}}$$

High $\rho(W)$ implies that pairwise erf-log attractions collectively produce a deeper cooperative potential well — the system becomes easier to destabilise coherently.

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§ III

Potential Energy Landscape

2.1 — Radial Spring (Confinement)

A harmonic well anchoring the system to equilibrium μ:

RADIAL SPRING ENERGY

\[\Phi_{\text{rad}}(X) = \frac{\alpha}{2} \sum_{i=1}^{N} \|x_i - \mu\|^2 \]

$\alpha = 0.10$	Radial spring constant; controls basin steepness

2.2 — Pairwise Interaction (erf-log Potential)

Pairs interact via an attraction–repulsion potential balancing cohesion against crowding:

The error-function attraction creates a finite-depth well at each pair; the logarithmic repulsion provides a hard-core floor that diverges as \(r_{ij}\to 0\).

PAIRWISE INTERACTION ENERGY

\[\Phi_{\text{pair}}(X) = \sum_{i < j} \Bigl[-\gamma\,\sigma\,\operatorname{erf}\!\Bigl(\frac{r_{i {\sigma}\Bigr) + \lambda\,\ln(r_{ij} + \varepsilon) \Bigr] \]

$\gamma = 1.00$	Coupling baseline — attraction amplitude
$\sigma = 1.00$	Attraction length-scale (erf inflection point)
$\lambda = 0.10$	Repulsion coefficient (logarithmic hard-core)
$\varepsilon = 10^{-5}$	Softening constant (prevents divergence at r = 0)

2.3 — Total Potential

FULL POTENTIAL

\[\Phi(X) = \Phi_{\text{rad}}(X) + \Phi_{\text{pair}}(X) = \frac{\alpha}{2}\sum_i \|x_i-\mu\|^2 + \sum_{i

USE INSTEAD OF SIMULATION

CLOSED-FORM MATRIX EXPRESSION — REPLACES O(N²) DOUBLE LOOP

Let \(D \in \mathbb{R}^{N\times N}\) be the Euclidean distance matrix: \(D_{ij} = \|x_i - x_j\|\). \[\Phi_{\text{rad}}(X) = \frac{\alpha}{2}\,\bigl\|X - \mathbf{1}\mu^\top\bigr\|_F^2 \] \[\Phi_{\text{pair}}(X) = \frac{1}

{2}\sum_{i\neq j}\!\left[-\gamma\sigma\operatorname{erf}\!\Bigl(\tfrac{D_{ij}}{\sigma}\Bigr) + \lambda\ln D_{ij} \right] = \mathbf{1}^\top\!\left[-\gamma\sigma\operatorname{erf}\!\Bigl(\tfrac{D}{\sigma}\Bigr) + \lambda\ln D\right]\!\mathbf{1}\Big/2 - \text{diag terms} \] \[\boxed{\Phi(X) = \frac{\alpha}{2}\|X - \mathbf{1}\mu^\top\|_F^2 - \frac{\gamma\sigma}{2}\left[\mathbf{1}^\top\operatorname{erf}\!\Bigl(\tfrac{D}{\sigma}\Bigr)\mathbf{1} - N\right] + \frac{\lambda}{2}\left[\mathbf{1}^\top\ln(D+\varepsilon I)\mathbf{1} - N\ln\varepsilon\right]} \] All operations are elementwise on \(D\); diagonal terms cancel by symmetry. Compute \(D\) once via \(D_{ij}^2 = \|x_i\|^2 + \|x_j\|^2 - 2x_i\cdot x_j\) (one matrix multiply).

Inputs needed: \(X \in \mathbb{R}^{N\times d}\), \(\mu \in \mathbb{R}^d\), parameters \(\alpha, \gamma, \sigma, \lambda, \varepsilon\)

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§ IV

Restoring Forces

3.1 — Radial (Gradient of Spring)

RADIAL FORCE ON AGENT I

\[F_i^{\text{rad}} = -\nabla_{x_i}\Phi_{\text{rad}} = -\alpha\,(x_i - \mu) \]

3.2 — Pairwise Force Decomposition

Differentiating the erf-log interaction with respect to \(x_i\):

PAIRWISE FORCE ON AGENT I FROM ALL J ≠ I

\[F_i^{\text{pair}} = \sum_{j \neq i} \underbrace{\left[\gamma\cdot\frac{2}{\sqrt{\pi}}\,e^{-(r_{ij}/\sigma)^2} - \frac{\lambda}{r_{ij}} \right]}_{\text{scalar

> radial coefficient}} \cdot \hat{r}_{ij} \] where $\hat{r}_{ij} = \dfrac{x_i - x_j}{r_{ij}}$ is the unit separation vector.

Note: The first term is the derivative of $\operatorname{erf}(r/\sigma)$ with respect to r, giving the Gaussian attraction kernel $\tfrac{2}{\sqrt{\pi}}e^{-r^2/\sigma^2}$.

3.3 — Total Force & Clamp

> TOTAL RESTORING FORCE
>
> $$F_i = F_i^{\text{rad}} + F_i^{\text{pair}}, \qquad \|F_i\| \leq F_{\max} = 100$$

USE INSTEAD OF SIMULATION

CLOSED-FORM FORCE MATRIX — REPLACES O(N²) AGENT LOOP

Define the scalar kernel matrix $K \in \mathbb{R}^{N\times N}$: $$K_{ij} = \frac{\gamma\cdot\frac{2}{\sqrt{\pi}}\,e^{-D_{ij}^2/\sigma^2} - \lambda/D_{ij}}{D_{ij}}, \quad i\neq j; \qquad K_{ii} = 0$$ Then the full force matrix is: $$\boxed{F = -\alpha(X - \mathbf{1}\mu^\top) + \left(K \odot \mathbf{1}\right)X - \operatorname{diag}(K\mathbf{1})\,X}$$ More explicitly, letting $\Delta_{ij} = x_i - x_j$: $$F_i^{\text{pair}} = \sum_{j\neq i} K_{ij}\,\Delta_{ij} = \Bigl(\sum_{j\neq i}K_{ij}\Bigr)x_i - \sum_{j\neq i}K_{ij}\,x_j = \bigl[(K\mathbf{1})_i\bigr]x_i - [KX]_i$$ $$\boxed{F^{\text{pair}} = \operatorname{diag}(K\mathbf{1})\,X - KX = \bigl(\operatorname{diag}(K\mathbf{1}) - K\bigr)X}$$ The matrix $L = \operatorname{diag}(K\mathbf{1}) - K$ is the *weighted graph Laplacian* of the force network. Total force in one expression: $$F = -\alpha(X - \mathbf{1}\mu^\top) - LX$$

Inputs needed: Distance matrix D, state X, equilibrium μ, parameters $\alpha,\gamma,\sigma,\lambda$

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§ V

Spectral Radius of the Hessian

The Hessian of Φ at X is approximated by a finite-difference directional second derivative, then the largest eigenvalue is extracted via randomised power iteration:

FINITE-DIFFERENCE HESSIAN-VECTOR PRODUCT

$$ H \cdot v \;\approx\; \frac{\Phi(X + h\,v) + \Phi(X - h\,v) - 2\,\Phi(X)}{h^2}\,v, \qquad h = 10^{-4} $$

Power iteration avoids explicitly forming the $Nd \times Nd$ Hessian. Each step costs two energy evaluations.

POWER ITERATION (K = 20 STEPS)

$$ v^{(k+1)} = \frac{H\,v^{(k)}}{\|H\,v^{(k)}\|}, \qquad \lambda_{\max} = \lim_{k\to K}\|H\,v^{(k)}\| $$

USE INSTEAD OF SIMULATION

CLOSED-FORM HESSIAN — REPLACES FINITE-DIFFERENCE POWER ITERATION

The Hessian of Φ decomposes analytically into radial and pairwise blocks. For the full system the $Nd\times Nd$ Hessian is: $$ \nabla^2_X \Phi = \alpha\, I_{Nd} + \frac{\partial^2 \Phi_{\text{pair}}}{\partial X^2} $$ The pairwise Hessian block between agents i and j ($i\neq j$) is a $d\times d$ matrix: $$ H_{ij}^{\text{pair}} = -\frac{\partial F_i^{\text{pair}}}{\partial x_j} = -K_{ij}\,I_d + \Bigl(\tfrac{\partial K_{ij}}{\partial r_{ij}}\Bigr)\frac{\Delta_{ij}\Delta_{ij}^\top}{D_{ij}} $$ where $\frac{\partial K_{ij}}{\partial r} = \frac{-4\gamma r}{\sqrt{\pi}\sigma^2}e^{-r^2/\sigma^2} + \frac{\lambda}{r^2}$ (second derivative of the pairwise potential kernel). **Analytical upper bound on $\lambda_{\max}$ — no iteration:** $$ \lambda_{\max}(\nabla^2\Phi) \leq \alpha + \max_i \sum_{j\neq i}\left|K_{ij}\right| + \left|\frac{\partial K_{ij}}{\partial r_{ij}}\right| \quad\text{(Gershgorin row-sum bound)} $$ **Critical manifold test without eigensolver:** $$ \lambda_{\max} > 1 \;\Longleftrightarrow\; \alpha + \max_i\sum_{j\neq i}\!\Bigl(\tfrac{2\gamma}{\sqrt\pi\sigma}e^{-D_{ij}^2/\sigma^2} + \tfrac{\lambda}{D_{ij}^2}\Bigr) > 1 $$ This is a single pass over the distance matrix — no simulation loop, no matrix factorisation.

Inputs needed:	Distance matrix D, parameters $\alpha,\gamma,\sigma,\lambda$ — purely from data snapshot X_t

CRITICAL MANIFOLD GEOMETRY

Critical Manifold $\mathcal{C}_{\text{man}}$

The system is **supercritical** — above the collapse threshold — when:

$$ \lambda_{\max}(H_\Phi(X)) > 1 $$

This is a level-set condition in configuration space. The critical manifold separates the basin of attraction from the unstable regime:

$$ \mathcal{C}_{\text{man}} = \bigl\{X \in \mathbb{R}^{N\times d} : \lambda_{\max}(\nabla^2_X\,\Phi) = 1\bigr\} $$

Fraction of trajectory above $\mathcal{C}_{\text{man}}$:

$$ \hat{p}_{\text{SC}} = \frac{1}{T}\sum_{t=1}^{T} \mathbf{1}\!\left[\lambda_{\max}^{(t)} > 1\right] $$

§ VI

Blind-Spot Detection Theory — Four Channels

BSDT is calibrated on a normal-period distribution $\mathcal{N}_0$ with mean μ_0 and covariance Σ_0. Each channel measures a distinct mode of deviation.

CHANNEL	FORMULA	GEOMETRIC INTERPRETATION
δ_C *Camouflage*	$\delta_C^{(i)} = (x_i-\mu_0)^\top \Sigma_0^{-1}(x_i-\mu_0)$	Squared Mahalanobis distance — ellipsoidal deviation from normal cloud
δ_G *Feature Gap*	$\delta_G^{(i)} = \|x_i-\mu_0\|^2 - \|V_k^\top(x_i-\mu_0)\|^2$	Residual energy in low-variance PCA directions — orthogonal complement of the principal subspace
δ_A *Activity Anomaly*	$\delta_A^{(i)} = \max\!\bigl(0,\;\|x_t^{(i)} - x_{t-1}^{(i)}\| - v_0\bigr)$	Excess velocity above normal-period 95th-percentile threshold v_0
δ_T *Temporal Novelty*	$\delta_T^{(i)} = \max\!\bigl(0,\,-\log p_{\text{KDE}}(x_t^{(i)})\bigr)$	Negative log-density under self-history KDE — measures temporal self-surprise

5.1 — Mahalanobis Ellipsoid (δ_C)

The iso-deviation surfaces are ellipsoids in $\mathbb{R}^d$ defined by Σ_0:

> MAHALANOBIS LEVEL SET AT RADIUS C
>
> $$\mathcal{E}_c = \bigl\{x \in \mathbb{R}^d : (x-\mu_0)^\top \Sigma_0^{-1}(x-\mu_0) = c^2\bigr\}$$ The axes of $\mathcal{E}_c$ are the eigenvectors of Σ_0, with half-lengths $c\sqrt{\lambda_k^{\Sigma}}$.

5.2 — PCA Decomposition (δ_G)

Let $V_k \in \mathbb{R}^{d\times k}$ be the top-k eigenvectors of Σ_0. The deviation decomposes as:

> PROJECTION DECOMPOSITION
>
> $$\|x_i - \mu_0\|^2 = \underbrace{\|V_k^\top(x_i-\mu_0)\|^2}_{\text{principal subspace}} + \underbrace{\delta_G^{(i)}}_{\text{gap (residual)}}$$ $$\delta_G^{(i)} = \|(I - V_k V_k^\top)(x_i - \mu_0)\|^2$$

5.3 — KDE Temporal Novelty (δ_T)

> KDE DENSITY ESTIMATE (GAUSSIAN KERNEL, BANDWIDTH H)
>
> $$\hat{p}_h(x) = \frac{1}{H}\sum_{s=1}^{H} \frac{1}{(2\pi h^2)^{d/2}} \exp\!\left(-\frac{\|x - x_{t-s}\|^2}{2h^2}\right)$$ $$\delta_T^{(i)} = \max\!\left(0,\; -\log\hat{p}_h(x_t^{(i)})\right)$$ Scott's rule: $h = H^{-1/(d+4)}$

5.4 — Total BSDT Score per Agent

> COMPOSITE SCORE
>
> $$\Delta^{(i)} = \delta_C^{(i)} + \delta_G^{(i)} + \delta_A^{(i)} + \delta_T^{(i)}$$

USE INSTEAD OF SIMULATION

CLOSED-FORM TOTAL BSDT ENERGY — REPLACES AGENT LOOP

$$ E_{\text{BSDT}}(X) = \sum_i \delta_C^{(i)} = \operatorname{tr}\!\bigl[(X-\mu_0\mathbf{1}^\top)\,\Sigma_0^{-1}(X-\mu_0\mathbf{1}^\top)^\top\bigr] $$ Equivalently using the centred matrix $\tilde X = X - \mathbf{1}\mu_0^\top$: $$ \boxed{E_{\text{BSDT}}(X) = \operatorname{tr}(\tilde X\,\Sigma_0^{-1}\tilde X^\top) = \|\tilde X\,\Sigma_0^{-1/2}\|_F^2} $$ One matrix multiply $\tilde X \Sigma_0^{-1}$, then a Frobenius norm squared. No loop over N agents.

Inputs needed: Current state X, calibrated μ_0, Σ_0^{-1} (fitted once on normal period)

5.5 — MFLS Signal (Frobenius gradient norm)

MFLS (MAHALANOBIS FIELD LINE SCORE)

$$ \text{MFLS}(X) = \left\|\nabla_X E_{\text{BSDT}}(X)\right\|_F = \left\|2\,(X - \mu_0\mathbf{1}^\top)\,\Sigma_0^{-1}\right\|_F $$ where $\nabla_{x_i} E = 2\,\Sigma_0^{-1}(x_i - \mu_0)$ for each agent i.

USE INSTEAD OF SIMULATION

CLOSED-FORM MFLS — SINGLE MATRIX OPERATION

$$ \boxed{\text{MFLS}(X) = 2\,\|\tilde X\,\Sigma_0^{-1}\|_F = 2\sqrt{\operatorname{tr}\!\bigl(\Sigma_0^{-1}\tilde X^\top \tilde X\,\Sigma_0^{-1}\bigr)}} $$ Pre-compute $A = \tilde X\,\Sigma_0^{-1}$ (one $N\times d$ matrix multiply using the stored inverse), then $\text{MFLS} = 2\|A\|_F$. No per-timestep loop, no gradient computation at runtime. **Optimal damping directly from MFLS:** $$ \gamma^*(X) = \frac{\tfrac{1}{4}\,\text{MFLS}^2}{\tfrac{1}{4}\,\text{MFLS}^2 + \theta} \qquad \bigl(\text{since } E_{\text{BSDT}} = \tfrac{1}{4}\,\text{MFLS}^2 \text{ only when }\Sigma_0^{-1}\text{ is symmetric}\bigr) $$ More precisely, using $E_{\text{BSDT}} = \|\tilde X\Sigma_0^{-1/2}\|_F^2$: $$ \boxed{\gamma^*(X) = \frac{\|\tilde X\Sigma_0^{-1/2}\|_F^2}{\|\tilde X\Sigma_0^{-1/2}\|_F^2 + \theta}} $$

Inputs needed: $\tilde X = X - \mathbf{1}\mu_0^\top$, pre-factored $\Sigma_0^{-1/2}$, cost θ

§ VII

Gradient Alignment Angle $\cos\theta$

This scalar measures the angular agreement between the BSDT energy gradient and the GravityEngine restoring force — a co-directional pair signals coherent collapse pressure:

COSINE ALIGNMENT

$$\cos\theta_t = \frac{\nabla_X E_{\text{BSDT}}(X_t) \cdot F(X_t)} {\|\nabla_X E_{\text{BSDT}}(X_t)\|\;\|F(X_t)\|} \in [-1, 1]$$

cos θ → +1	BSDT gradient and physical force fully aligned — mutual reinforcement of collapse
cos θ → 0	Orthogonal — BSDT detects hidden risk not visible to physical potential
cos θ → −1	Anti-aligned — forces oppose; stabilising competition

USE INSTEAD OF SIMULATION

CLOSED-FORM COS θ — DIRECT MATRIX EXPRESSION

Substituting the closed-form gradient $\nabla_X E = 2\tilde X\Sigma_0^{-1}$ and force $F = -\alpha\tilde X_\mu - LX$: $$\cos\theta_t = \frac{\operatorname{tr}\!\bigl((\tilde X_t\Sigma_0^{-1})^\top F_t\bigr)} {\|\tilde X_t\Sigma_0^{-1}\|_F\;\|F_t\|_F} = \frac{\operatorname{tr}(\Sigma_0^{-1}\tilde X_t^\top F_t)} {\|\tilde X_t\Sigma_0^{-1}\|_F\;\|F_t\|_F}$$ where $F_t = -\alpha(X_t - \mathbf{1}\mu^\top) - L_t X_t$ (from §IV Laplacian form). In terms of the precomputed matrices $A_t = \tilde X_t\Sigma_0^{-1}$ and F_t: $$\boxed{\cos\theta_t = \frac{\langle A_t,\, F_t\rangle_F} {\|A_t\|_F\;\|F_t\|_F}}$$ Two Frobenius inner products on $N\times d$ matrices — no per-agent, per-timestep iteration.

Inputs needed: Pre-computed $A_t = \tilde X_t\Sigma_0^{-1}$ (from MFLS step) and F_t (from force step)

§ VIII

Optimal Adaptive Damping \(\gamma^*\)

The optimal intervention intensity balances BSDT blind-spot energy against a cost parameter \(\theta\):

> OPTIMAL DAMPING POLICY
>
> \[\gamma^*(X_t) = \frac{E_{\text{BSDT}}(X_t)}{E_{\text{BSDT}}(X_t) + \theta}, \qquad \theta = 1 \text{ (intervention cost)} \]

\(\gamma^* \in [0,1)\) is a smooth saturation function of blind-spot energy — it approaches 1 as risk diverges and is 0 when the system is at normal-period baseline.

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§ IX

CCyB Calibration & Welfare Loss

8.1 — Countercyclical Capital Buffer

> CCYB MAPPING (SCALED TO BASEL III RANGE)
>
> \[\text{CCyB}(t) = 250 \cdot \frac{\gamma^*(t)\,\sigma_\ell(t)/\kappa}{\max_s[\gamma^*(s)\,\sigma_\ell(s)/\kappa]} \quad \in [0,\,250]\text{ bps} \] where \(\sigma_\ell(t)\) is the cross-sectional standard deviation of leverage at time \(t\).

USE INSTEAD OF SIMULATION

CLOSED-FORM CCYB FROM DATA — NO TRAJECTORY SIMULATION

Substituting the closed-form \(\gamma^*\) and noting \(\kappa\) cancels in the ratio: \[\text{CCyB}(t) = 250\cdot\frac{E_t\,\sigma_\ell(t)/(E_t+\theta)}{\max_s\bigl[E_s\,\sigma_\ell(s)/(E_s+\theta)\bigr]} \] where \(E_t = \|\tilde X_t\,\Sigma_0^{-1/2}\|_F^2\) computed directly from the data panel. In full: \[\boxed{\text{CCyB}(t) = 250\cdot\frac{\operatorname{tr}(\tilde X_t\Sigma_0^{-1}\tilde X_t^\top)\cdot\sigma_\ell(t)} {\max_s\!\left[\operatorname{tr}(\tilde X_s\Sigma_0^{-1}\tilde X_s^\top)\cdot\sigma_\ell(s)\cdot \frac{1}{\operatorname{tr}(\tilde X_s\Sigma_0^{-1}\tilde X_s^\top)+\theta}\right] \cdot\bigl(\operatorname{tr}(\tilde

$$X_t\Sigma_0^{-1}\tilde X_t^\top)+\theta\bigr)}}$$ **Practical shortcut:** Let $e_t = \operatorname{tr}(\tilde X_t\Sigma_0^{-1}\tilde X_t^\top)$. Then: $$\text{CCyB}(t) = 250\cdot\frac{e_t\,\sigma_\ell(t)\,(e_t+\theta)^{-1}} {\max_s\bigl[e_s\,\sigma_\ell(s)\,(e_s+\theta)^{-1}\bigr]}$$ Compute the vector $\{e_t\}$ across all T timesteps with a single batched trace operation, then apply the ratio. No agent-level simulation required.

Inputs needed: Panel $\{X_t\}$, calibrated μ_0, Σ_0^{-1}, leverage std series $\{\sigma_\ell(t)\}$, θ

8.2 — Consumption-Equivalent Welfare Loss

WELFARE LOSS FORMULA

$$\mathcal{W} = 100\!\left(1 - \exp\!\left[-\frac{1}{\sigma\,\theta\,T_w}\sum_{t=1}^{T_w} \bigl(\Phi(X_t) - \Phi^{\text{cf}}(X_t)\bigr) \right]\right) \;\text{pp}$$ where Φ^{cf} is the counterfactual energy under optimal γ^*.

USE INSTEAD OF SIMULATION

CLOSED-FORM WELFARE LOSS — COUNTERFACTUAL ENERGY FROM FORMULA

The counterfactual energy Φ^{cf} is the energy the system would have if forces were damped by $\gamma^*(t)$. Under optimal damping the pairwise coupling is scaled by $(1-\gamma^*)$, giving: $$\Phi^{\text{cf}}(X_t) = \Phi_{\text{rad}}(X_t) + (1-\gamma^*(t))\,\Phi_{\text{pair}}(X_t)$$ Therefore the energy gap is purely pairwise: $$\Phi(X_t) - \Phi^{\text{cf}}(X_t) = \gamma^*(t)\,\Phi_{\text{pair}}(X_t) = \frac{E_t}{E_t+\theta}\cdot\Phi_{\text{pair}}(X_t)$$ Substituting into the welfare formula: $$\boxed{\mathcal{W} = 100\!\left(1 - \exp\!\left[-\frac{1}{\sigma\,\theta\,T_w}\sum_{t=1}^{T_w} \frac{e_t}{e_t+\theta}\cdot\Phi_{\text{pair}}(X_t) \right]\right)}$$ where $e_t = \|\tilde X_t\Sigma_0^{-1/2}\|_F^2$ and Φ_{pair} uses the closed-form matrix expression from §III. The entire welfare loss is computable from the data panel and calibrated parameters alone — no simulation run required.

Inputs needed: Crisis-window panel $\{X_t\}_{t=1}^{T_w}$, calibrated μ_0,Σ_0^{-1}, parameters $\sigma,\theta,\gamma,\lambda$

T_w	Crisis window length in quarters (default 10)
σ, θ	Calibration parameters (elasticity, intervention cost)
pp	Percentage points of consumption-equivalent loss

§ X

Geometric & Trigonometric Summary of Collapse

System collapse can be read as a sequence of geometric events in the configuration manifold $\mathbb{R}^{N\times d}$, beginning with the network losing its independence structure and ending with full intervention:

COLLAPSE EVENT SEQUENCE — NETWORK TO INTERVENTION

$$ \begin{array}{ll} 0. & \text{Network synchronises}: \quad \rho(W) \nearrow N \;\text{(loss of independence)} \\[6pt] 1. & X_t \text{ exits the normal ellipsoid } \mathcal{E}_c: \quad \delta_C^{(i)} > c^2 \\[6pt] 2. & X_t \text{ crosses the critical manifold}: \quad \lambda_{\max}(\nabla^2\Phi) > 1 \\[6pt] 3. & \text{MFLS spikes}: \quad \|\nabla_X E_{\text{BSDT}}\|_F \gg 0 \\[6pt] 4. & \text{Gradient alignment}: \quad \cos\theta_t \to 1 \;\text{(coherent collapse)} \\[6pt] 5. & \gamma^*(t) \to 1 \;\text{(full intervention warranted)} \end{array} $$

TWO SPECTRAL RADII — NETWORK VS. ENERGY LANDSCAPE

$$ \underbrace{\rho(W)}_{\text{network}}\quad\text{vs.}\quad\underbrace{\lambda_{\max}(\nabla^2\Phi)}_{\text{Hessian}} $$

$$ \rho(W) \nearrow N \;\Rightarrow\; \text{contagion topology tightens (macro)} $$

$$ \lambda_{\max}(\nabla^2\Phi) > 1 \;\Rightarrow\; \text{energy landscape goes unstable (micro)} $$

Together they form a dual early-warning pair: the network radius signals *who* is coupled; the Hessian radius signals *whether* the coupled state is structurally stable.

ELLIPSOID AXIS LENGTHS (MAHALANOBIS GEOMETRY)

$$ \text{Semi-axis}_k = c\sqrt{\lambda_k^{\Sigma_0}}, \quad k = 1,\ldots,d $$ where $\lambda_k^{\Sigma_0}$ are eigenvalues of the normal-period covariance.

ANGLE OF DEVIATION FROM PRINCIPAL SUBSPACE

$$ \sin^2\!\phi_i = \frac{\delta_G^{(i)}}{\|x_i - \mu_0\|^2} = 1 - \frac{\|V_k^\top(x_i-\mu_0)\|^2}{\|x_i-\mu_0\|^2} $$ $\phi_i = 0$ when agent i stays within the principal subspace; $\phi_i = \pi/2$ when deviation is entirely orthogonal to it.

SHRINKAGE GEOMETRY — INTERPOLATION BETWEEN TWO COVARIANCE STRUCTURES

$$ \hat\Sigma(\rho) = (1-\rho)\,S + \rho\,\frac{\operatorname{tr}(S)}{N}I, \qquad \rho \in [0,1] $$ At $\rho=0$: full empirical covariance (max observed coupling). At $\rho=1$: spherical identity (complete independence prior). The shrinkage path traces a straight line in matrix space between these two geometries.

USE INSTEAD OF SIMULATION

MASTER PIPELINE — COMPLETE SYSTEM OUTPUTS FROM DATA ALONE

Given panel $\{X_t\}_{t=0}^T$ and calibrated parameters $(\mu_0,\Sigma_0,\mu,\alpha,\gamma,\sigma,\lambda,\theta)$, compute all outputs in order:

\[\begin{array}{lll} \textbf{Step} & \textbf{Formula} & \textbf{Output} \\[4pt] \hline\\[-6pt] 1 & \tilde X_t = X_t - \mathbf{1}\mu_0^\top & \text{centred state}\\[6pt] 2 & D_{ij}^{(t)} = \|x_i^{(t)}-x_j^{(t)}\| & \text{distance matrix (1 matmul)}\\[6pt] 3 & e_t = \|\tilde X_t\Sigma_0^{-1/2}\|_F^2 & E_{\text{BSDT}} \text{ (1 matmul + trace)}\\[6pt] 4 & \text{MFLS}_t = 2\|\tilde X_t\Sigma_0^{-1}\|_F & \text{field line score}\\[6pt] 5 & \gamma^*_t = e_t/(e_t+\theta) & \text{optimal damping (scalar)}\\[6pt] 6 & \Phi_t = \frac{\alpha}{2}\|X_t-\mathbf{1}\mu^\top\|_F^2 + \mathbf{1}^\top\![-\gamma\sigma\operatorname{erf}(D/\sigma)+\lambda\ln D]\mathbf{1}/2 & \text{total energy}\\[6pt] 7 & \lambda_{\max}^{\text{bound}} = \alpha + \max_i\sum_j(\tfrac{2\gamma}{\sqrt\pi\sigma}e^{-D_{ij}^2/\sigma^2}+\tfrac{\lambda}{D_{ij}^2}) & \text{critical manifold test}\\[6pt] 8 & \tilde\rho_t = 1 + (N-1)\bar W_{\text{off},t} & \text{network radius proxy}\\[6pt] 9 & \text{CCyB}(t) = 250\cdot e_t\sigma_\ell(t)(e_t+\theta)^{-1}/\max_s[\cdot] & \text{capital buffer (bps)}\\[6pt] 10 &

\mathcal{W} = 100(1-e^{-\frac{1}{\sigma\theta T_w}\sum_t\gamma^*_t\Phi_{\text{pair},t}}) & \text{welfare loss (pp)} \end{array} \]

Steps 1–10 require **no simulation loop**, no ODE integration, no iterative eigensolver. Each step is a closed matrix or scalar operation on the observed panel.

`One-time fit:` \(\mu_0, \Sigma_0^{-1}, \Sigma_0^{-1/2}, V_k\) — computed once on normal-period data, then stored

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§ XI

Complete Closed-Form Representation

This section consolidates every analytical formula in the system into a single self-contained reference. No simulation, no iteration, no numerical integration — each quantity is a direct algebraic function of the observed panel \(\{X_t\}\) and a small set of parameters fitted once on normal-period data.

11.1 — Calibration Objects (fit once, store permanently)

NORMAL-PERIOD STATISTICS — FIT ON X_NORMAL ∈ $\mathbb{R}^{T_0 \times N \times D}$

\[\mu_0 = \frac{1}{T_0 N}\sum_{t,i} x_t^{(i)} \in \mathbb{R}^d \qquad\text{(global mean)} \] \[\Sigma_0 = \frac{1}{T_0 N}\sum_{t,i}(x_t^{(i)}-\mu_0)(x_t^{(i)}-\mu_0)^\top + 10^{-8}I \in \mathbb{R}^{d\times d} \] \[\Sigma_0^{-1},\quad \Sigma_0^{-1/2} \;\text{(Cholesky or eigendecomposition — computed once)} \] \[V_k \in \mathbb{R}^{d\times k} \;\text{— top-}k\text{ eigenvectors of }\Sigma_0\text{ (PCA basis)} \] \[v_0 = \operatorname{Pctl}_{95}\!\bigl\{\|x_t^{(i)} - x_{t-1}^{(i)}\| : t=1,\ldots,T_0,\; i=1,\ldots,N\bigr\} \quad\text{(velocity threshold)} \]

T_0	Length of normal (calibration) period
k	Number of PCA components retained (default 4)
v_0	95th-percentile velocity in normal period

11.2 — Intermediate Quantities (per snapshot X_t)

A — CENTRED STATE MATRIX

$$ \tilde{X}_t = X_t - \mathbf{1}_N \mu_0^\top \in \mathbb{R}^{N\times d} $$

B — EUCLIDEAN DISTANCE MATRIX (ONE MATRIX MULTIPLY)

$$ D_{ij}^{(t)} = \left\|x_i^{(t)} - x_j^{(t)}\right\|_2, \qquad D^2 = \mathbf{q}\mathbf{1}^\top + \mathbf{1}\mathbf{q}^\top - 2X_t X_t^\top $$

where $\mathbf{q}_i = \|x_i\|^2$. Set $D_{ii} = 0$, add softening ε where needed.

C — MAHALANOBIS PROJECTION (ONE MATRIX MULTIPLY)

$$ A_t = \tilde{X}_t\,\Sigma_0^{-1} \in \mathbb{R}^{N\times d} $$

D — KERNEL WEIGHT MATRIX FOR PAIRWISE FORCES

$$ K_{ij} = \frac{1}{D_{ij}}\left[\frac{2\gamma}{\sqrt{\pi}}\,e^{-D_{ij}^2/\sigma^2} - \frac{\lambda}{D_{ij}}\right], \quad i\neq j;\qquad K_{ii}=0 $$

E — GRAPH LAPLACIAN OF FORCE NETWORK

$$ L_t = \operatorname{diag}(K_t\,\mathbf{1}) - K_t \in \mathbb{R}^{N\times N} $$

11.3 — Primary Outputs (closed-form)

FIFTEEN CLOSED-FORM EXPRESSIONS

① RADIAL SPRING ENERGY

$$ \Phi_{\text{rad}}(X_t) = \frac{\alpha}{2}\left\|X_t - \mathbf{1}_N\mu^\top\right\|_F^2 $$

② PAIRWISE INTERACTION ENERGY

$$ \Phi_{\text{pair}}(X_t) = \frac{1}{2}\left[-\gamma\sigma\,\mathbf{1}^\top\operatorname{erf}\!\Bigl(\tfrac{D_t}{\sigma}\Bigr)\mathbf{1} +\lambda\,\mathbf{1}^\top\ln(D_t+\varepsilon)\,\mathbf{1} + N\gamma\sigma - N\lambda\ln\varepsilon \right] $$

③ TOTAL POTENTIAL ENERGY

$$ \Phi(X_t) = \Phi_{\text{rad}}(X_t) + \Phi_{\text{pair}}(X_t) $$

④ TOTAL RESTORING FORCE MATRIX

$$ F_t = -\alpha\bigl(X_t - \mathbf{1}_N\mu^\top\bigr) - L_t\,X_t \in \mathbb{R}^{N\times d} $$

⑤ BSDT BLIND-SPOT ENERGY (MAHALANOBIS CHANNEL Δ_C)

$$ e_t = E_{\text{BSDT}}(X_t) = \left\|\tilde{X}_t\,\Sigma_0^{-1/2}\right\|_F^2 = \operatorname{tr}\!\bigl(\tilde{X}_t\,\Sigma_0^{-1}\tilde{X}_t^\top\bigr) $$

⑥ FEATURE GAP ENERGY (Δ_G)

$$ \delta_G(X_t) = \left\|\tilde{X}_t\right\|_F^2 - \left\|\tilde{X}_t V_k\right\|_F^2 = \operatorname{tr}\!\bigl(\tilde{X}_t^\top\tilde{X}_t\bigr) - \operatorname{tr}\!\bigl(V_k^\top\tilde{X}_t^\top\tilde{X}_t V_k\bigr) $$

⑦ ACTIVITY ANOMALY (Δ_A, REQUIRES TWO CONSECUTIVE SNAPSHOTS)

$$ \delta_A(X_t,X_{t-1}) = \left\|\max\!\bigl(0,\;\|X_t - X_{t-1}\|_{\text{row}} - v_0\bigr)\right\|_1 $$ where $\|\cdot\|_{\text{row}}$ denotes the vector of row-wise norms.

⑧ MFLS — MAHALANOBIS FIELD LINE SCORE

$$ \text{MFLS}(X_t) = 2\left\|A_t\right\|_F = 2\left\|\tilde{X}_t\,\Sigma_0^{-1}\right\|_F $$

⑨ GRADIENT ALIGNMENT ANGLE

$$ \cos\theta_t = \frac{\langle A_t,\; F_t\rangle_F}{\left\|A_t\right\|_F\;\left\|F_t\right\|_F} = \frac{\operatorname{tr}(A_t^\top F_t)}{\left\|A_t\right\|_F\;\left\|F_t\right\|_F} $$

⑩ CRITICAL MANIFOLD TEST (GERSHGORIN BOUND — NO EIGENSOLVER)

$$ \lambda_{\max}(\nabla^2\Phi) \;\leq\; \alpha + \max_{i}\sum_{j\neq i}\!\left[\frac{2\gamma}{\sqrt{\pi}\,\sigma}\,e^{-D_{ij}^2/\sigma^2} + \frac{\lambda}{D_{ij}^2} \right] $$ $$ \text{Supercritical} \iff \text{above bound} > 1 $$

⑪ OPTIMAL ADAPTIVE DAMPING

$$ \gamma^*(X_t) = \frac{e_t}{e_t + \theta} $$

⑫ LEDOIT-WOLF SHRINKAGE INTENSITY (CLOSED-FORM, NO OPTIMISATION)

$$ \rho = \operatorname{clip}\!\left(\frac{\left\|X_c^{\odot 2}\right\|_F^2\big/(T^2 N^2) - \|S\|_F^2/(TN^2)} {\|S\|_F^2 - [\operatorname{tr}(S)]^2/N},\;0,\;1\right) $$

⑬ NETWORK SPECTRAL RADIUS PROXY (NO EIGENDECOMPOSITION)

$$ \tilde\rho(W_t) = 1 + (N-1)\,\bar{W}_{\text{off},t}, \qquad \bar{W}_{\text{off}} = \frac{\mathbf{1}^\top W_t\mathbf{1} - N}{N(N-1)} $$

⑭ CCYB IN BASIS POINTS

$$ \text{CCyB}(t) = 250\cdot\frac{e_t\,\sigma_\ell(t)\,(e_t+\theta)^{-1}}{\displaystyle\max_{s\in[1,T]}\!\bigl[e_s\,\sigma_\ell(s)\,(e_s+\theta)^{-1}\bigr]} $$

⑮ CONSUMPTION-EQUIVALENT WELFARE LOSS

$$ \mathcal{W} = 100\!\left(1 - \exp\!\left[-\frac{1}{\sigma\,\theta\,T_w} \sum_{t=1}^{T_w} \frac{e_t}{e_t+\theta}\cdot\Phi_{\text{pair}}(X_t) \right]\right) \;\text{pp} $$

11.4 — Dependency Graph

Every output above depends on at most three upstream quantities. The full dependency order is:

COMPUTATION DAG — LEFT TO RIGHT, NO CYCLES

$$ \underbrace{X_t}_{\text{data}} \;\longrightarrow\; \begin{cases} \tilde X_t \;\longrightarrow\; A_t \;\longrightarrow\; e_t \;\longrightarrow\; \gamma^*_t \;\longrightarrow\; \text{CCyB},\;\mathcal{W} \\[4pt] \tilde X_t \;\longrightarrow\; A_t \;\longrightarrow\; \text{MFLS}_t \\[4pt] \tilde X_t \;\longrightarrow\; \delta_G \\[4pt] D_t \;\longrightarrow\; K_t \;\longrightarrow\; L_t \;\longrightarrow\; F_t \;\longrightarrow\; \cos\theta_t \\[4pt] D_t \;\longrightarrow\; \Phi_{\text{pair}} \;\longrightarrow\; \Phi_t,\;\mathcal{W} \\[4pt] D_t \;\longrightarrow\; \lambda_{\max}^{\text{bound}} \;\longrightarrow\; \mathcal{C}_{\text{man}} \\[4pt] W_t \;\longrightarrow\; \tilde\rho_t \end{cases} $$

11.5 — Complete Parameter Reference

SYMBOL	VALUE	ROLE	APPEARS IN
α	0.10	Radial spring constant	①③④⑩
γ	1.00	Pairwise attraction amplitude	②③④⑩

SYMBOL	VALUE	ROLE	APPEARS IN
σ	1.00	Attraction length-scale	②③④⑩⑮
λ	0.10	Logarithmic repulsion coefficient	②③④⑩
ε	10^{-5}	Distance softening	②③
θ	1.00	Intervention cost	⑪⑭⑮
κ	1.00	CCyB scaling	⑭ (cancels)
k	4	PCA components retained	⑥
v_0	Pctl_{95}	Normal-period velocity threshold	⑦
T_w	10	Crisis window (quarters)	⑮
$F_{\max}$	100	Force clamp (applied post ④)	④

11.6 — What the Closed Form Eliminates

SIMULATION PROCEDURE	REPLACED BY	COST REDUCTION
$O(N^2)$ pairwise energy loop	Elementwise ops on D_t + vector sums	$O(N^2 d) \to O(N^2)$
$O(N^2)$ pairwise force loop	Laplacian matrix multiply $L_t X_t$	$O(N^2 d) \to O(N^2 + Nd)$
$K=20$ Hessian power iterations	Gershgorin row-sum over D_t	20 energy evals $\to$ 1 matrix pass
Per-agent BSDT energy loop	$\operatorname{tr}(\tilde X_t \Sigma_0^{-1} \tilde X_t^\top)$	$O(Nd) \to$ 1 matmul + trace
Per-agent gradient loop (MFLS)	$2\|\tilde X_t \Sigma_0^{-1}\|_F$	$O(Nd^2) \to$ 1 matmul + norm

SIMULATION PROCEDURE	REPLACED BY	COST REDUCTION
Full eigensolver for $\rho(W)$	$1 + (N-1)\bar W_{\text{off}}$	$O(N^3) \to O(N^2)$
LW shrinkage optimisation loop	Direct ratio of Frobenius norms	Iterative $\to$ 3 scalar ops
Counterfactual trajectory simulation	$\gamma^*_t \cdot \Phi_{\text{pair}}(X_t)$	Full re-simulation $\to$ scalar multiply
Per-timestep $\cos\theta$ loop	$\langle A_t, F_t\rangle_F / (\|A_t\|_F\|F_t\|_F)$	Loop over T $\to$ 1 inner product

§ XII

The Unified Energy Field — MFLS as Vector

The system already has four channels from BSDT and one scalar from MFLS. The upgrade: treat MFLS not as a magnitude floating free of direction, but as the norm of a full gradient field. That field is built from three composable energy forms — Quadsurf, Expogate, and Signed LR — each capturing a distinct mode of instability. Together they form the complete blind-spot energy E_{BS}.

12.1 — The Channel State Vector

> BSDT STATE VECTOR — INPUT TO ALL THREE ENERGY FORMS
>
> $$S = \bigl(\delta_C,\;\delta_G,\;\delta_A,\;\delta_T\bigr)^\top \in \mathbb{R}^4$$ Each component is a non-negative scalar measuring a distinct deviation mode. $S = 0$ at normal-period baseline; $\|S\|$ grows as the system departs.

12.2 — Three Energy Forms

> ① QUADSURF — QUADRATIC CURVATURE ENERGY

$$ E_{\text{quad}}(S) = S^\top A\, S, \qquad A \in \mathbb{R}^{4\times 4},\;\; A = A^\top \succ 0 $$ $$ \nabla_S E_{\text{quad}} = 2A S $$

A	Symmetric positive-definite coupling matrix — encodes pairwise channel interactions; off-diagonals capture how one channel amplifies another
STAS	Generalised Mahalanobis distance in channel space; reduces to $\|S\|^2$ when $A = I$
Geometry	Iso-energy surfaces are ellipsoids in $\mathbb{R}^4$; axes are eigenvectors of A with radii $\propto \lambda_k(A)^{-1/2}$

② EXPOGATE — EXPONENTIAL THRESHOLD ENERGY

$$ E_{\text{exp}}(S) = \sum_{k=1}^{4} \exp\!\bigl(\beta_k\,S_k\bigr), \qquad \beta_k > 0 $$ $$ \nabla_S E_{\text{exp}} = \bigl(\beta_k\,e^{\beta_k S_k}\bigr)_{k=1}^4 $$

β_k	Per-channel gate rate — controls how steeply energy grows near threshold; large β_k makes channel k a hair-trigger detector
Geometry	Energy grows super-exponentially near threshold; gradient component explodes, pulling u sharply toward the triggering channel

③ SIGNED LR — LINEAR DIRECTIONAL ENERGY

$$ E_{\text{lr}}(S) = w^\top S, \qquad w \in \mathbb{R}^4 $$ $$ \nabla_S E_{\text{lr}} = w \quad\text{(constant — independent of } S\text{)} $$

w	Weight vector encoding directional bias — $w_k > 0$ means channel k drives toward instability; $w_k < 0$ means stabilising
Geometry	Energy level sets are parallel hyperplanes; gradient is a fixed vector regardless of system state — a persistent directional pull

12.3 — Unified Blind-Spot Energy

FULL COMPOSITE ENERGY

$$ E_{\text{BS}}(S) = \underbrace{S^\top A S}_{\text{curvature}} + \underbrace{\textstyle\sum_k e^{\beta_k S_k}}_{\text{explosion}} + \underbrace{w^\top S}_{\text{bias}} $$

UNIFIED GRADIENT — THE COLLAPSE DIRECTION VECTOR

$$ \nabla_S E_{\text{BS}} = \underbrace{2AS}_{\text{structural}} + \underbrace{\bigl(\beta_k e^{\beta_k S_k}\bigr)_k}_{\text{critical transition}} + \underbrace{w}_{\text{directional}} $$

The three terms are additive and independent. Each fires in its own regime: Signed LR is always present, Quadsurf grows with displacement, Expogate dominates near threshold.

12.4 — MFLS as the Magnitude of the Field

MFLS — SCALAR INTENSITY FROM VECTOR NORM

$$ \text{MFLS}(S) = \left\|\nabla_S E_{\text{BS}}\right\| = \left\|2AS + \bigl(\beta_k e^{\beta_k S_k}\bigr)_k + w\right\| $$

UNIT COLLAPSE DIRECTION

$$ u = \frac{\nabla_S E_{\text{BS}}}{\text{MFLS}} \in \mathbb{R}^4, \qquad \|u\| = 1 $$

u points toward the fastest-growing instability in channel space. It is a closed-form vector computable directly from S, A, β, and w.

12.5 — Per-Channel Collapse Attribution

CHANNEL ATTRIBUTION — FRACTION OF TOTAL MFLS^2 PER CHANNEL

$$ c_k = \frac{\bigl[\nabla_S E_{\text{BS}}\bigr]_k^2}{\text{MFLS}^2} \in [0,1], \qquad \sum_{k=1}^4 c_k = 1 $$

$$ \bigl[\nabla_S E_{\text{BS}}\bigr]_k = 2[AS]_k + \beta_k\,e^{\beta_k S_k} + w_k $$

The dominant channel $k^* = \arg\max_k c_k$

identifies which BSDT mode is driving collapse — directly readable from the gradient without any simulation.

12.6 — Reduction to Original BSDT

SPECIAL CASE: QUADSURF ONLY, $A = \Sigma_0^{-1}$, $S = (X - M_0)$

$$E_{\text{BS}}(S)\big|_{A=\Sigma_0^{-1},\,\beta=0,\,w=0} = (x-\mu_0)^\top\Sigma_0^{-1}(x-\mu_0) = \delta_C$$

$$\text{MFLS}\big|_{\text{single agent}} = \|2\Sigma_0^{-1}(x-\mu_0)\|$$

The original BSDT Mahalanobis formulation is exactly Quadsurf with $A = \Sigma_0^{-1}$, confirming the unified energy as a strict generalisation of what was built.

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§ XIII

The Control Law — Force Decomposition & Minimal Damping

The system dynamics are not stopped — only the collapse-directed component of the force is damped. Everything orthogonal to the gradient runs free. This is the principle of minimal intervention: the geometry of the energy landscape determines which part of the force is dangerous, and only that part is attenuated.

13.1 — Force Decomposition

Given the total restoring force $F \in \mathbb{R}^{N \times d}$ from the GravityEngine (§IV) and the unit collapse direction $u \in \mathbb{R}^4$ from the channel gradient (§XII.4), decompose:

PARALLEL COMPONENT — COLLAPSE-DIRECTED, POTENTIALLY DANGEROUS

$$F_\parallel = \langle F,\, u\rangle\, u, \qquad \langle F,\,u\rangle = \operatorname{tr}(F^\top u)$$

PERPENDICULAR COMPONENT — ORTHOGONAL TO COLLAPSE, LOCALLY NEUTRAL

$$ F_\perp = F - F_\parallel = F - \langle F,\, u\rangle\, u $$ $$ \langle F_\perp,\, u\rangle = 0 \quad \text{(exact, by construction)} $$

PYTHAGOREAN IDENTITY — ENERGY PARTITION

$$ \|F\|^2 = \|F_\parallel\|^2 + \|F_\perp\|^2 = \langle F,u\rangle^2 + \|F - \langle F,u\rangle u\|^2 $$

13.2 — Why Perpendicular is Neutral (Proof)

FIRST-ORDER ENERGY CHANGE ALONG ANY DIRECTION V

$$ \frac{d}{dt}E_{\text{BS}}\Big|_{v} = \nabla_S E_{\text{BS}} \cdot v = \text{MFLS} \cdot \langle u, v\rangle $$ For $v = F_\perp$: $\langle u, F_\perp\rangle = 0$, therefore $\dfrac{d}{dt}E_{\text{BS}}\big|_{F_\perp} = 0$. Moving along $F_\perp$ does not change instability to first order — the system stays on the same iso-instability surface.

SECOND-ORDER CORRECTION — CURVATURE OF THE ISO-SURFACE

$$ \frac{d^2}{dt^2}E_{\text{BS}}\Big|_{F_\perp} = F_\perp^\top\,\nabla^2_S E_{\text{BS}}\,F_\perp = F_\perp^\top\!\bigl(2A + \operatorname{diag}(\beta_k^2 e^{\beta_k S_k})\bigr)F_\perp $$ This is zero only when $F_\perp$ lies in the null space of $\nabla^2 E_{\text{BS}}$. Otherwise the iso-surface curves back — the second-order growth rate is bounded by $\lambda_{\max}(\nabla^2 E_{\text{BS}})\cdot\|F_\perp\|^2$. This is exactly what the curvature threshold $\lambda_{\max} > 1$ detects.

13.3 — The Control Law

CONTROLLED DYNAMICS — GENERAL FORM

$$ \frac{dX}{dt} = F_\perp + (1 - \gamma^*)\,F_\parallel = F - \gamma^*\,\langle F,u\rangle\,u $$

FULLY CLOSED FORM — ALL TERMS SUBSTITUTED

$$\boxed{ \frac{dX}{dt} = F_t - \frac{e_t}{e_t + \theta}\cdot \frac{\langle F_t,\;\nabla_S E_{\text{BS}}(S_t)\rangle}{\|\nabla_S E_{\text{BS}}(S_t)\|^2} \cdot\nabla_S E_{\text{BS}}(S_t) }$$

where $e_t = \|\tilde X_t\Sigma_0^{-1/2}\|_F^2$, $\;F_t = -\alpha(X_t - \mathbf{1}\mu^\top) - L_t X_t$, and $\nabla_S E_{\text{BS}} = 2AS + (\beta_k e^{\beta_k S_k})_k + w$.

$\gamma^* \to 0$	Far from collapse: $dX/dt \approx F_t$ — control is invisible, system evolves freely
$\gamma^* \to 1$	At collapse: $dX/dt \approx F_\perp$ — only safe tangential motion permitted
$\gamma^* = \frac{1}{2}$	Half-energy state: parallel force halved, perpendicular untouched

13.4 — Domain-Universal Form

The control law $\dot X = F - \gamma^*\langle F,u\rangle u$ holds in any domain where S, E_{BS}, and F can be defined:

DOMAIN TABLE — SAME LAW, DIFFERENT INTERPRETATIONS

$$\begin{array}{lll} \textbf{Domain} & \textbf{State } X & \textbf{Dynamics } F \\[4pt] \hline\\[-4pt] \text{Financial systemic risk} & \text{leverage, positions, correlations} & \text{GravityEngine force field} \\[5pt] \text{Fluid dynamics} & \text{velocity modes, vorticity} & \text{Navier-Stokes operator} \\[5pt] \text{Neural networks} & \text{activations, gradient norms} & \text{backpropagation update step} \\[5pt] \text{Epidemiology} & \text{S/I/R compartment fractions} & \text{transmission rate dynamics} \end{array}$$

The structure is not domain-specific. It is a geometric control law defined entirely by the energy landscape.

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§ XIV

The $\tan(\theta)$ Collapse Condition

The gradient gives direction; the Hessian gives curvature. The angle θ between the trajectory and the principal instability direction is the exact bridge between geometry, energy intensity, and the spectral threshold $\lambda_{\max}$. When $\tan\theta$ crosses a critical value, collapse is inevitable regardless of other system properties.

14.1 — The Trajectory Angle

DEFINITION OF θ — ANGLE BETWEEN MOTION AND COLLAPSE DIRECTION

$$ \theta = \angle\!\bigl(\dot X,\; u\bigr) = \arccos\!\left(\frac{\langle \dot X,\, u\rangle}{\|\dot X\|}\right) $$

$$ \cos\theta = \frac{\|F_\parallel\|}{\|F\|} = \frac{|\langle F, u\rangle|}{\|F\|}, \qquad \sin\theta = \frac{\|F_\perp\|}{\|F\|} $$

TANGENT — THE COLLAPSE-TO-SAFETY RATIO

$$ \tan\theta = \frac{\|F_\perp\|}{\|F_\parallel\|} = \frac{\bigl\|F - \langle F,u\rangle u\bigr\|}{\bigl|\langle F,u\rangle\bigr|} = \sqrt{\frac{\|F\|^2}{\langle F,u\rangle^2} - 1} $$

$\tan\theta \to 0$	Trajectory fully collapse-directed — all force drives toward instability manifold
$\tan\theta \to \infty$	Trajectory fully perpendicular — motion is tangential, locally safe
$\tan\theta = 1$	Equal components — critical balance at θ = 45°; the natural intervention threshold

14.2 — Connection to Gradient Alignment $\cos\theta_t$

The gradient alignment angle $\cos\theta_t$ from §VII is exactly the cosine of this angle. Therefore:

TAN θ IN TERMS OF THE EXISTING COS θ_T MEASUREMENT

$$ \tan\theta = \frac{\sin\theta_t}{\cos\theta_t} = \sqrt{\frac{1 - \cos^2\theta_t}{\cos^2\theta_t}} $$

$$ = \tan\!\left(\arccos\!\left(\frac{\langle\nabla_X E_{\text{BS}},\; F\rangle}{\|\nabla_X E_{\text{BS}}\|\,\|F\|}\right)\right) $$

No new computation is required — $\tan\theta$ is a direct transform of the already-computed $\cos\theta_t$.

14.3 — Connection to Curvature and $\lambda_{\max}$

TAN Θ IN TERMS OF MFLS AND SPECTRAL STRUCTURE

Let v_1 be the principal eigenvector of $\nabla^2 E_{\text{BS}}$ (direction of maximum curvature), and ϕ the angle between ∇E_{BS} and v_1:

$$\tan\theta_{\text{curv}} = \frac{\text{MFLS}\cdot\sin\phi}{\lambda_{\max}(\nabla^2 E_{\text{BS}})\cdot\cos\phi} = \frac{\text{MFLS}}{\lambda_{\max}}\cdot\tan\phi$$

When gradient aligns with principal curvature direction ($\phi \to 0$): $\tan\theta_{\text{curv}} \to 0$ — collapse is curvature-dominated. When orthogonal ($\phi \to \pi/2$): curvature is irrelevant to the collapse direction.

14.4 — The Exact Collapse Condition

CRITICAL MANIFOLD GEOMETRY

Three Equivalent Collapse Conditions

SPECTRAL (HESSIAN OF GRAVITYENGINE)

$$\lambda_{\max}\!\bigl(\nabla^2_X\Phi(X_t)\bigr) > 1$$

ANGULAR (TRAJECTORY GEOMETRY)

$$\tan\theta_t < \tan\theta^* \qquad\text{where}\quad \tan\theta^* = \frac{\sqrt{\|F\|^2 - \text{MFLS}^2\cdot[\gamma^*]^2}}{\text{MFLS}\cdot\gamma^*}$$

ENERGETIC (CURVATURE OVERWHELMS GRADIENT)

$$\lambda_{\max}\!\bigl(\nabla^2_S E_{\text{BS}}(S_t)\bigr) > \frac{\text{MFLS}^2}{e_t} = \frac{\|\nabla_S E_{\text{BS}}\|^2}{\|\tilde X_t\Sigma_0^{-1/2}\|_F^2}$$

All three conditions are equivalent at the critical manifold $\mathcal{C}_{\text{man}}$. The angular condition is geometric (observable); the spectral is algebraic (eigenvalue); the energetic is closed-form testable (ratio of norms).

14.5 — Hessian of the Unified Energy

∇²E_BS — EXACT CLOSED FORM

$$\nabla^2_S E_{\text{BS}} = \underbrace{2A}_{\text{Quadsurf}} + \underbrace{\operatorname{diag}\!\bigl(\beta_k^2\,e^{\beta_k S_k}\bigr)}_{\text{Expogate}} + \underbrace{0}_{\text{Signed LR (linear, no curvature)}}$$

SPECTRAL RADIUS OF THE UNIFIED HESSIAN (CLOSED-FORM BOUND)

$$\lambda_{\max}(\nabla^2 E_{\text{BS}}) \leq 2\lambda_{\max}(A) + \max_k\!\bigl(\beta_k^2\,e^{\beta_k S_k}\bigr)$$

The Expogate diagonal grows exponentially near threshold — it can overwhelm the Quadsurf structural curvature $2\lambda_{\max}(A)$ when $\beta_k S_k \gg 1$. This is the instability explosion mechanism encoded in closed form.

14.6 — The Complete Four-Layer Chain

DETECT → MEASURE → LOCATE → CONTROL

$$\underbrace{S_t = \bigl(\delta_C,\delta_G,\delta_A,\delta_T\bigr)^\top}_{\text{① Detect (BSDT)}} \;\longrightarrow\; \underbrace{e_t,\;\text{MFLS}_t}_{\text{② Measure (intensity)}} \;\longrightarrow\; \underbrace{u_t = \tfrac{\nabla_S E_{\text{BS}}}{\text{MFLS}_t}}_{\text{③ Locate (direction)}} \;\longrightarrow\; \underbrace{\dot X = F_t - \gamma^*_t\langle F_t,u_t\rangle u_t}_{\text{④ Control (damp)}}$$

COLLAPSE GEOMETRY — THE TAN(θ) BRIDGE

> $$\underbrace{\tan\theta_t}_{\substack{\text{trajectory}\\\text{angle}}} \;\longleftrightarrow\; \underbrace{\frac{\text{MFLS}_t}{\lambda_{\max}(\nabla^2 E_{\text{BS}})}}_{\substack{\text{gradient-to-curvature}\\\text{ratio}}} \;\longleftrightarrow\; \underbrace{\lambda_{\max}(\nabla^2\Phi_t)}_{\substack{\text{Hessian of}\\\text{GravityEngine}}}$$ The $\tan\theta$ formula is the geometric lens through which the gradient direction (MFLS), the curvature of the energy landscape (Hessian), and the spectral collapse threshold ($\lambda_{\max} > 1$) all become the same statement read three different ways.

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§ XV

The Master Operator — One Equation

Everything built across the preceding fourteen sections — BSDT channels, unified energy, Quadsurf, Expogate, Signed LR, MFLS, spectral radius, critical manifold, adaptive damping, force decomposition, tan(θ) geometry — compresses into a single operator. It is stated in full, then unpacked layer by layer.

15.1 — Construction: Eight Steps to One Equation

STEP	SYMBOL	EXPRESSION	SOURCE
(A)	S_t	$\mathcal{B}(X_t) = \bigl(\delta_C,\,\delta_G,\,\delta_A,\,\delta_T\bigr)^\top$	BSDT mapping §VI
(B)	E_{BS}	$S_t^\top A S_t + \sum_k e^{\beta_k S_{k,t}} + w^\top S_t$	Unified energy §XII
(C)	g_t	$\nabla_S E_{\text{BS}}(S_t) = 2AS_t + (\beta_k e^{\beta_k S_{k,t}})_k + w$	Collapse direction §XII
(D)	MFLS_t	$\|g_t\|$	Magnitude §XII.4
(E)	u_t	$g_t / \|g_t\|$	Unit direction §XII.4

STEP	SYMBOL	EXPRESSION	SOURCE
(F)	F_t	$-\alpha(X_t - \mathbf{1}\mu^\top) - L_t X_t$	GravityEngine §IV
(G)	e_t	$\|\tilde X_t\,\Sigma_0^{-1/2}\|_F^2$	BSDT energy §VI
(H)	γ_t^*	$e_t\,/\,(e_t + \theta)$	Adaptive damping §VIII

15.2 — The Master Equation

FIFTEEN CLOSED-FORM EXPRESSIONS

THE BSDT MASTER OPERATOR

$$\boxed{ \dot{X}_t \;=\; \underbrace{F_t}_{\text{free dynamics}} \;-\; \underbrace{\frac{e_t}{e_t + \theta}}_{\gamma_t^*} \;\cdot\; \underbrace{\frac{\langle F_t,\; g_t\rangle}{\|g_t\|^2}}_{\text{projection scalar}} \;\cdot\; \underbrace{g_t}_{\nabla_S E_{\text{BS}}(\mathcal{B}(X_t))} }$$

where $g_t = \nabla_S E_{\text{BS}}\!\bigl(\mathcal{B}(X_t)\bigr) = 2A\mathcal{B}(X_t) + \bigl(\beta_k e^{\beta_k [\mathcal{B}(X_t)]_k}\bigr)_k + w$

EQUIVALENT COMPACT FORM

$$\dot{X}_t = F_t - \gamma_t^*\,\langle F_t,\,u_t\rangle\,u_t, \qquad u_t = \frac{g_t}{\|g_t\|}$$

EQUIVALENT SPLIT FORM

\[\dot{X}_t = \underbrace{F_t - \langle F_t,\,u_t\rangle\,u_t}_{F_\perp\;\text{(always free)}} \;+\; \underbrace{(1-

\gamma_t^*)\,\langle F_t,\,u_t\rangle\,u_t}_{(1-\gamma^*)F_\parallel\;\text{(damped)}} \]

15.3 — Anatomy of the Operator

The master equation has exactly three factors in its second term. Each factor carries a distinct geometric meaning:

FACTOR 1 — ADAPTIVE FRICTION SCALAR Γ^*

$$ \gamma_t^* = \frac{e_t}{e_t+\theta} \in [0,1) $$ Measures how far the system is from normal baseline. Zero at rest, approaching 1 at collapse. All the BSDT blind-spot machinery — Mahalanobis distance, covariance inversion, normal-period calibration — is compressed into this single scalar.

FACTOR 2 — PROJECTION SCALAR $\langle F_T, G_T \rangle / \|G_T\|^2$

$$ \frac{\langle F_t,\,g_t\rangle}{\|g_t\|^2} = \frac{\|F_t\|\cos\theta_t}{\|g_t\|} $$ The signed magnitude of the force component along the collapse direction, normalised by energy gradient strength. This is the $\cos\theta_t$ measurement from §VII, scaled by $\|F_t\|/\|g_t\|$. Positive means force drives toward collapse; negative means force drives away.

FACTOR 3 — COLLAPSE DIRECTION VECTOR G_T

$$ g_t = \underbrace{2AS_t}_{\text{structural (Quadsurf)}} + \underbrace{(\beta_k e^{\beta_k S_{k,t}})_k}_{\text{threshold (Expogate)}} + \underbrace{w}_{\text{bias (Signed LR)}} $$ Points toward the fastest-growing instability in channel space. All three MFLS forms — curvature, explosion, direction — are compressed into this single vector.

15.4 — What the Operator Contains

SYSTEM COMPONENT	WHERE IT LIVES IN THE OPERATOR
BSDT 4-channel detection	$\mathcal{B}(X_t)$ — argument of g_t
MFLS scalar	$\Vert g_t\Vert$ — norm of the gradient
Quadsurf curvature	$2AS_t$ — first term of g_t
Expogate threshold	$(\beta_k e^{\beta_k S_{k,t}})_k$ — second term of g_t
Signed LR direction	w — third term of g_t
Critical manifold $\mathcal{C}_\text{man}$	Implicitly: $\lambda_\text{max}(\nabla^2 E_\text{BS}) > \text{MFLS}^2/e_t \Leftrightarrow \gamma^* \to 1$
GravityEngine potential	$F_t = -\alpha\tilde X_{t,\mu} - L_t X_t$ — the free dynamics term
Gradient alignment $\cos\theta_t$	$\langle F_t, g_t\rangle / (\Vert F_t\Vert\Vert g_t\Vert)$ — the projection scalar (normalised)
$\tan\theta$ collapse condition	$\langle F_t, u_t\rangle / \Vert F_t\Vert$ embedded in the projection; collapse when this dominates
Ledoit-Wolf network	$L_t = \text{diag}(K_t\mathbf{1}) - K_t$ inside F_t
Adaptive damping γ^*	Leading scalar factor — all BSDT energy in one number
Welfare calibration	$\gamma^*_t \cdot \Phi_\text{pair}(X_t)$ — the controlled energy reduction summed over T_w

15.5 — Behaviour at the Limits

> NORMAL REGIME — FAR FROM COLLAPSE
>
> $$ e_t \approx 0 \;\Rightarrow\; \gamma_t^* \approx 0 \;\Rightarrow\; \dot X_t \approx F_t $$ The control term vanishes. The system evolves under its own dynamics without interference. BSDT is watching but not acting.

> CRITICAL REGIME — AT THE COLLAPSE MANIFOLD

$$ e_t \gg \theta \;\Rightarrow\; \gamma_t^* \approx 1 \;\Rightarrow\; \dot X_t \approx F_t - \frac{\langle F_t,g_t\rangle}{\|g_t\|^2}\,g_t = F_\perp $$

Only the perpendicular (safe) component survives. The parallel (collapse-directed) force is fully removed. The system is confined to the tangent plane of the instability surface.

PERPENDICULAR TRAJECTORY — SAFE MOTION AT ANY ENERGY LEVEL

$$ \langle F_t, g_t\rangle = 0 \;\Rightarrow\; \dot X_t = F_t \;\text{ regardless of } \gamma_t^* $$

When the natural dynamics are already perpendicular to the collapse direction, the operator does nothing. The control is genuinely minimal — it never damps what is safe.

15.6 — Universal Instantiation

THE OPERATOR IN FOUR DOMAINS — SAME EQUATION, FOUR INTERPRETATIONS

$$ \begin{array}{llll} \textbf{Domain} & X_t & F_t & \mathcal{B}(X_t) \\[4pt] \hline\\[-4pt] \text{Systemic risk} & \text{leverage matrix} & {-\alpha\tilde X - LX} & (\delta_C,\delta_G,\delta_A,\delta_T) \\[5pt] \text{Fluid dynamics} & \text{velocity field} & \mathcal{N}[X_t] & \text{modal energy channels} \\[5pt] \text{Neural network} & \text{weight tensor} & {-\nabla_W \mathcal{L}} & \text{gradient anomaly channels} \\[5pt] \text{Epidemiology} & (S,I,R,\ldots) & \beta SI/N,\,\ldots & \text{compartment deviation channels} \end{array} $$

15.7 — Fully Expanded Closed Form

Substituting every term back to raw data and calibrated parameters — no functions left undefined:

USE INSTEAD OF SIMULATION

COMPLETE CLOSED-FORM EXPANSION OF THE MASTER OPERATOR

$$ \dot X_t = \underbrace{\bigl[-\alpha(X_t - \mathbf{1}\mu^\top) - L_t X_t\bigr]}_{F_t} \;-\; \underbrace{\frac{\|\tilde X_t\Sigma_0^{-1/2}\|_F^2}{\|\tilde X_t\Sigma_0^{-1/2}\|_F^2 + \theta}}_{\gamma_t^*} \;\cdot\; \frac{\displaystyle\Bigl\langle F_t,\;2A\mathcal{B}(X_t) + \bigl(\beta_k e^{\beta_k[\mathcal{B}(X_t)]_k}\bigr)_k + w\Bigr\rangle} $$

$${\displaystyle\bigl\|2A\mathcal{B}(X_t) + \bigl(\beta_k e^{\beta_k[\mathcal{B}(X_t)]_k}\bigr)_k + w\bigr\|^2} \;\cdot\; \Bigl[2A\mathcal{B}(X_t) + \bigl(\beta_k e^{\beta_k[\mathcal{B}(X_t)]_k}\bigr)_k + w\Bigr]$$

Every symbol is either observed data (X_t), fitted once on normal-period data ($\mu_0, \Sigma_0, A, V_k, v_0$), or a scalar parameter ($\alpha, \gamma, \sigma, \lambda, \theta, \beta_k, w$). The operator is a pure algebraic function of the current state snapshot.

`Inputs at time t:` X_t only — all other quantities are constants pre-computed at calibration

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§ XVI

Lyapunov Stability Certificate — Corrected & Complete

The master operator damps the collapse-directed force — but does it actually guarantee stability? A Lyapunov certificate proves that the control law causes E_{BS} to decrease along controlled trajectories. This section gives the rigorous proof, including the chain-rule step that was previously missing: $g_t \in \mathbb{R}^4$ lives in channel space, while $\dot X_t \in \mathbb{R}^{N\times d}$ lives in state space. The two are connected by the Jacobian of the BSDT map $\mathcal{B}$.

16.1 — The Lyapunov Candidate

LYAPUNOV FUNCTION

$$V(X_t) = E_{\text{BS}}\!\bigl(\mathcal{B}(X_t)\bigr) = e_t \;\geq\; 0$$ $$V(X_t) = 0 \iff \mathcal{B}(X_t) = 0 \iff X_t = \mu_0\,\mathbf{1}^\top \quad\text{(all agents at normal-period mean)}$$ V is non-negative, zero exactly at the normal-period baseline, radially unbounded ($V \to \infty$ as $\|X_t\| \to \infty$), and smooth in X_t. It is a valid global Lyapunov candidate.

16.2 — The Chain Rule Gap (The Error)

The naive computation writes $\dot V = g_t \cdot \dot X_t$, treating g_t and $\dot X_t$ as living in the same space. They do not. The correct chain rule is:

CORRECT CHAIN RULE — THROUGH THE BSDT JACOBIAN

$$\frac{dV}{dt} = \frac{d}{dt}E_{\text{BS}}(\mathcal{B}(X_t)) = \nabla_S E_{\text{BS}}\big|_{S_t} \cdot \frac{d\mathcal{B}(X_t)}{dt} = g_t^\top \underbrace{\frac{\partial \mathcal{B}}{\partial X}\bigg|_{X_t}}_{\mathbf{J}_t \;\in\;\mathbb{R}^{4\times Nd}} \operatorname{vec}(\dot X_t)$$

where $\mathbf{J}_t = \partial\mathcal{B}/\partial X\big|_{X_t} \in \mathbb{R}^{4\times Nd}$ is the Jacobian of the BSDT map and $\operatorname{vec}(\dot X_t)$ stacks $\dot X_t$ into a column vector. Defining the *pullback gradient*:

$$\tilde g_t = \mathbf{J}_t^\top g_t \in \mathbb{R}^{Nd} \quad\Longleftrightarrow\quad \tilde G_t = \operatorname{mat}(\tilde g_t) \in \mathbb{R}^{N\times d}$$

the chain rule becomes simply:

$$\frac{dV}{dt} = \langle \tilde G_t,\; \dot X_t\rangle_F$$

$\tilde G_t$ is the gradient of V *in state space* — it pulls the channel-space gradient g_t back through the BSDT Jacobian into $\mathbb{R}^{N\times d}$.

16.3 — The BSDT Jacobian $\mathbf{J}_t$

Each BSDT channel is a function of X_t. The four rows of $\mathbf{J}_t$ are:

ROW 1 — CAMOUFLAGE CHANNEL Δ_C

$$\frac{\partial \delta_C}{\partial X_t} = 2\tilde X_t\,\Sigma_0^{-1} \in \mathbb{R}^{N\times d}$$

ROW 2 — FEATURE GAP CHANNEL Δ_G

$$\frac{\partial \delta_G}{\partial X_t} = 2\tilde X_t\,(I - V_k V_k^\top) \in \mathbb{R}^{N\times d}$$

ROW 3 — ACTIVITY ANOMALY Δ_A

$$\frac{\partial \delta_A}{\partial X_t} = \frac{X_t - X_{t-1}}{\|X_t - X_{t-1}\|_{\text{row}}} \odot \mathbf{1}[\|X_t - X_{t-1}\|_{\text{row}} > v_0] \in \mathbb{R}^{N\times d}$$

(row-wise normalised velocity gradient, zero where below threshold)

ROW 4 — TEMPORAL NOVELTY Δ_T

$$ \frac{\partial \delta_T}{\partial X_t} = -\frac{1}{\hat p_h(X_t)}\frac{\partial \hat p_h}{\partial X_t} = \frac{1}{h^2}\sum_{s}\frac{e^{-\|X_t - X_{t-s}\|^2/2h^2}}{\sum_{s'}e^{-\|X_t-X_{t-s'}\|^2/2h^2}}(X_t - X_{t-s}) \in \mathbb{R}^{N\times d} $$

FULL PULLBACK GRADIENT (STATE-SPACE GRADIENT OF V)

$$ \tilde G_t = [g_t]_1\,\frac{\partial\delta_C}{\partial X_t} + [g_t]_2\,\frac{\partial\delta_G}{\partial X_t} + [g_t]_3\,\frac{\partial\delta_A}{\partial X_t} + [g_t]_4\,\frac{\partial\delta_T}{\partial X_t} $$ This is a weighted sum of the four channel Jacobians, weighted by the components of the collapse direction g_t. It tells you: for each agent i and each dimension j, how much does a small change in $x_t^{(i,j)}$ increase the total blind-spot energy?

16.4 — Time Derivative Under Control (Correct)

DV/DT — COMPLETE DERIVATION

$$ \frac{dV}{dt} = \langle\tilde G_t,\;\dot X_t\rangle_F = \left\langle\tilde G_t,\; F_t - \gamma_t^*\,\frac{\langle F_t, g_t\rangle}{\|g_t\|^2}\,g_t\right\rangle_F $$ $$ = \langle\tilde G_t, F_t\rangle_F - \gamma_t^*\,\frac{\langle F_t, g_t\rangle}{\|g_t\|^2}\,\langle\tilde G_t, g_t\rangle_F $$ Now: $\langle\tilde G_t, g_t\rangle_F = \|\tilde G_t\|\,\|g_t\|\,\cos\psi_t$ where ψ_t is the angle between the state-space pullback $\tilde G_t$ and the channel-space direction g_t (after embedding). In the special case where $\delta_G = \delta_A = \delta_T = 0$ (pure Mahalanobis regime): $$ \tilde G_t = 2[g_t]_1\,\tilde X_t\Sigma_0^{-1}, \qquad \langle\tilde G_t, g_t\rangle_F = 2[g_t]_1\,\langle\tilde X_t\Sigma_0^{-1}, g_t\rangle_F $$ In general: $$ \boxed{ \frac{dV}{dt} = \langle\tilde G_t, F_t\rangle_F\left(1 - \gamma_t^*\,\frac{\langle F_t, g_t\rangle\,\langle\tilde G_t, g_t\rangle_F}{\langle F_t, g_t\rangle\,\|g_t\|^2}\right) } $$ which simplifies to the compact form: $$ \boxed{\frac{dV}{dt} = \langle\tilde G_t, F_t\rangle_F - \gamma_t^*\,\frac{\langle F_t, g_t\rangle}{\|g_t\|^2}\,\langle\tilde G_t, g_t\rangle_F} $$

16.5 — When the Previous Result Was Correct

The simplified result $\dot V = \langle g_t, F_t\rangle(1-\gamma^*)$ from the earlier version holds when $\tilde G_t \propto g_t$ — i.e. when the state-space gradient is parallel to the channel-space gradient. This is exactly the pure Mahalanobis case (δ_C only, $A = \Sigma_0^{-1}$) where:

> CONDITION FOR EQUALITY OF THE TWO FORMS
>
> $$\tilde G_t = \frac{2}{\|g_t\|}\,g_t\,\|\tilde X_t\Sigma_0^{-1}\|_F \iff \text{pure }\delta_C\text{ regime, }g_t\text{ aligned with }\tilde G_t$$ In the full four-channel system with Quadsurf + Expogate + Signed LR, $\tilde G_t$ and g_t are generically \emph{not} parallel, and the full expression must be used.

16.6 — The Stability Theorem (Corrected)

CRITICAL MANIFOLD GEOMETRY

Lyapunov Stability Theorem — Correct Form

Define the **two inner products**:

$$P_t = \langle\tilde G_t,\; F_t\rangle_F \quad\text{(state-space: force against pullback gradient)}$$ $$Q_t = \langle F_t,\; g_t\rangle \cdot \langle\tilde G_t,\; g_t\rangle_F \quad\text{(cross-space: projection coupling)}$$

Then:

$$\frac{dV}{dt} = P_t - \frac{\gamma_t^*}{\|g_t\|^2}\,Q_t$$

The controlled system is **Lyapunov stable** ($\dot V \leq 0$) when:

$$\boxed{P_t \leq \frac{\gamma_t^*}{\|g_t\|^2}\,Q_t}$$

Three regimes:

`P_t < 0`	Force opposes pullback gradient — system is self-stabilising regardless of control; $\dot V < 0$ always
`P_t > 0, Q_t > 0`	Force drives toward instability, but control projects it away — stable iff $\gamma^* Q_t \geq \|g_t\|^2 P_t$

`P_t > 0, Q_t < 0`	Both uncontrolled drift and projection term are destabilising — control alone cannot stabilise; $\dot V > 0$ always

CRITICAL CONTROL STRENGTH FOR STABILISATION

$$\gamma_{\min}^* = \frac{\|g_t\|^2\,P_t}{Q_t} \quad\text{(when }P_t > 0,\,Q_t > 0\text{)}$$ If $\gamma_t^* > \gamma_{\min}^*$, the control is sufficient. The BSDT energy $\gamma^* = e_t/(e_t+\theta)$ must exceed $\gamma_{\min}^*$ — this gives a direct constraint on the intervention cost parameter θ: $$\theta < e_t\!\left(\frac{Q_t}{\|g_t\|^2 P_t} - 1\right)$$ This is the critical θ condition: if the intervention cost is too high, the control is too weak to prevent collapse.

16.7 — Stability Margin (Corrected)

TRUE STABILITY MARGIN

$$\mathcal{M}_t = \frac{\gamma_t^* Q_t/\|g_t\|^2 - P_t}{\|P_t\| + \|Q_t\|/\|g_t\|^2}$$ $\mathcal{M}_t > 0$: controlled system is stable. $\mathcal{M}_t < 0$: control is insufficient. $\mathcal{M}_t = 0$: the boundary — the control is exactly at the stability threshold. This is the correct replacement for $\mathcal{M}_t = -\cos\theta_t$, which only holds in the pure Mahalanobis single-channel case.

MFLS DECAY RATE (CORRECTED)

$$\frac{d\,\text{MFLS}_t}{dt} = -\frac{1}{\|g_t\|}\left(P_t - \frac{\gamma_t^*}{\|g_t\|^2}\,Q_t\right) = \frac{\mathcal{M}_t\,(\|P_t\| + \|Q_t\|/\|g_t\|^2)}{\|g_t\|}$$ The MFLS decays when $\mathcal{M}_t > 0$ and grows when $\mathcal{M}_t < 0$. The rate of decay/growth is proportional to the stability margin and inversely proportional to the current MFLS magnitude — large MFLS slows its own decay rate.

§ XVII

The Early Warning Composite Score

The system produces five independent collapse signals: MFLS, $\lambda_{\max}(\nabla^2\Phi)$, $\rho(W)$, γ^*, and $\cos\theta_t$. Each captures a different facet — intensity, curvature, network coupling, energy saturation, and angular alignment. The composite early warning score fuses all five into a single number with a natural $[0,1]$ scale and a proven threshold.

17.1 — Normalised Signal Components

FIVE NORMALISED COLLAPSE SIGNALS

$$ \begin{array}{lll} \xi_1 &= \gamma_t^* = e_t/(e_t+\theta) & \text{energy saturation} \in [0,1) \\[5pt] \xi_2 &= \min\!\bigl(\lambda_{\max}(\nabla^2\Phi_t),\,1\bigr) & \text{spectral criticality} \in [0,1] \\[5pt] \xi_3 &= \bigl(\tilde\rho(W_t) - 1\bigr)/(N-1) = \bar W_{\text{off},t} & \text{network synchrony} \in [0,1] \\[5pt] \xi_4 &= \max\!\bigl(0,\,\cos\theta_t\bigr) & \text{alignment (clamped positive)} \in [0,1] \\[5pt] \xi_5 &= 1 - 1/(1 + \text{MFLS}_t) & \text{MFLS saturation} \in [0,1) \end{array} $$

17.2 — The Early Warning Score

COMPOSITE EWS — WEIGHTED GEOMETRIC MEAN

$$ \text{EWS}_t = \prod_{k=1}^{5} \xi_k^{w_k}, \qquad \sum_k w_k = 1,\quad w_k > 0 $$

The geometric mean is used rather than arithmetic because the signals are multiplicative risk factors: if any one signal is zero, the system cannot be at the edge of collapse.

EQUAL-WEIGHT CLOSED FORM

$$ \text{EWS}_t\big|_{w_k = 1/5} = \left[\gamma_t^* \cdot \min(\lambda_{\max}^{\Phi},1) \cdot \bar W_{\text{off}} \cdot \max(0,\cos\theta_t) \cdot \frac{\text{MFLS}_t}{1+\text{MFLS}_t}\right]^{1/5} $$

COLLAPSE THRESHOLD

$$\text{Collapse alert} \iff \text{EWS}_t > \text{EWS}^* = \prod_k \xi_k^{*(w_k)}$$

where ξ_k^* are the threshold values of each signal at the critical manifold. At $\mathcal{C}_{\text{man}}$: $\lambda_{\max}^{\Phi} = 1$, $\gamma^* = e^*/(e^*+\theta)$, giving EWS^* as a computable calibration constant from normal-period data.

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§ XVIII

Collapse Time Prediction — Analytical Escape Time

Given the current state X_t and velocity $\dot X_t$, how many steps until the system crosses the critical manifold? This is the escape time problem — solved analytically using the energy gradient and the current rate of change.

18.1 — Linear Extrapolation (First Order)

FIRST-ORDER ESCAPE TIME ESTIMATE

$$\tau_{\text{lin}} = \frac{e^* - e_t}{\dot e_t}, \qquad \dot e_t = \frac{dV}{dt} = \langle g_t, F_t\rangle(1-\gamma_t^*)$$

where e^* is the energy level at the critical manifold (calibrated from normal-period data as the e_t value corresponding to $\lambda_{\max}^{\Phi} = 1$). When $\dot e_t > 0$ (energy growing), τ_{lin} gives the time to threshold.

18.2 — Curvature-Corrected Estimate (Second Order)

SECOND-ORDER ESCAPE TIME — PARABOLIC CORRECTION

$$\ddot e_t = g_t^\top \nabla^2_S E_{\text{BS}}\, g_t \cdot \|\dot X_t\|^2 + \langle g_t, \ddot X_t\rangle$$

$$\tau_{\text{quad}} = \frac{-\dot e_t + \sqrt{\dot e_t^2 + 2\ddot e_t(e^* - e_t)}}{\ddot e_t}$$

This accounts for the acceleration of energy growth — critical near threshold where the Expogate term causes super-exponential acceleration.

18.3 — The Collapse Horizon

> COLLAPSE HORIZON — TIME WITHIN WHICH INTERVENTION MUST OCCUR
>
> $$ \tau_{\text{safe}} = \tau_{\text{lin}} - \frac{1}{\alpha + \lambda_{\max}(L_t)} $$
>
> The subtracted term is the characteristic relaxation time of the GravityEngine. If $\tau_{\text{safe}} < 0$, the system is already past the point where unassisted stabilisation is possible — the control must be engaged immediately. If $\tau_{\text{safe}} > 0$, there is a window for intervention.

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§ XIX

Stochastic Extension — Noise, Itô Correction, Fokker-Planck

The deterministic system is an idealisation. Real financial systems, physical systems, and biological systems are driven by noise. Adding a diffusion term $\sigma_n\,dW_t$ (Wiener process) fundamentally changes the collapse geometry — the critical manifold acquires a stochastic halo, and the energy itself has an Itô correction.

19.1 — Stochastic Dynamics

> STOCHASTIC CONTROLLED SDE
>
> $$ dX_t = \underbrace{\bigl(F_t - \gamma_t^*\langle F_t,u_t\rangle u_t\bigr)}_{\text{controlled drift}}\,dt + \sigma_n\,dW_t $$ where W_t is an $N\times d$ matrix Wiener process and $\sigma_n > 0$ is the noise amplitude.

19.2 — Itô Correction to Energy

> STOCHASTIC ENERGY DYNAMICS (ITÔ'S LEMMA)
>
> $$ de_t = \underbrace{\langle g_t, F_t\rangle(1-\gamma_t^^*)\,dt}_{\text{deterministic drift}} + \underbrace{\frac{\sigma_n^2}{2}\operatorname{tr}(\nabla^2_X E_{\text{BS}})\,dt}_{\text{Itô correction}} + \underbrace{\sigma_n\,\langle g_t,\,dW_t\rangle}_{\text{noise term}} $$ The Itô correction \(\frac{\sigma_n^2}

{2}\operatorname{tr}(\nabla^2_X E_{\text{BS}})\) is always positive — noise systematically \emph{inflates} energy on average, pushing the system toward the critical manifold even when the deterministic drift is stabilising.

NOISE-ADJUSTED STABILITY CONDITION

$$ \text{Stable in expectation} \iff \langle g_t, F_t\rangle(1-\gamma_t^*) < -\frac{\sigma_n^2}{2}\operatorname{tr}(\nabla^2_X E_{\text{BS}}) $$

The deterministic stabilising force must overcome the noise injection. The Itô correction $\frac{\sigma_n^2}{2}\operatorname{tr}(2A + \operatorname{diag}(\beta_k^2 e^{\beta_k S_k})) = \sigma_n^2\bigl(\operatorname{tr}(A) + \sum_k \beta_k^2 e^{\beta_k S_k}/2\bigr)$ grows near threshold — making collapse more likely under noise exactly when it is most dangerous.

19.3 — Fokker-Planck Equation

PROBABILITY DENSITY EVOLUTION OVER THE ENERGY LANDSCAPE

$$ \frac{\partial p}{\partial t}(e,t) = -\frac{\partial}{\partial e}\!\Bigl[\mu_e(e)\,p\Bigr] + \frac{\sigma_n^2}{2}\frac{\partial^2 p}{\partial e^2} $$

$$ \mu_e(e) = \langle g, F\rangle(1-\gamma^*(e)) + \frac{\sigma_n^2}{2}\operatorname{tr}(\nabla^2 E_{\text{BS}}) $$

The stationary distribution (if it exists) satisfies:

$$ p^*(e) \propto \exp\!\left(-\frac{2}{\sigma_n^2}\int^e \mu_e(e')\,de'\right) $$

This is the probability density of the energy level — the distribution of where the system spends its time. A heavy tail toward high e signals elevated systemic risk even in the "normal" regime.

19.4 — Stochastic Critical Manifold

PROBABILITY OF CROSSING C_MAN WITHIN TIME T

$$ P(\tau_{\text{cross}} < \tau) \approx 1 - \exp\!\left(-\frac{\tau}{\tau_{\text{Kramers}}}\right) $$

$$ \tau_{\text{Kramers}} = \frac{2\pi}{\sqrt{|\mu_e'(e_t)|\cdot|\mu_e'(e^*)|}}\cdot\exp\!\left(\frac{2(e^* - e_t)}{\sigma_n^2}\right) $$

Kramers' escape time formula: the mean time to cross the critical manifold grows exponentially with the energy gap $(e^* - e_t)/\sigma_n^2$. Noise reduces this time by shrinking the effective barrier. Monitoring this ratio gives a probabilistic collapse horizon.

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§ XX

The Inverse Problem — Parameter Identification from Collapse

The forward problem is: given parameters, predict collapse. The inverse problem is: given an observed collapse trajectory, what parameters caused it? This is the identification problem — it tells you not just that collapse happened, but \emph{which} structural features of the system were responsible.

20.1 — The Observable

OBSERVED TRAJECTORY RESIDUAL

$$ R_t(\theta_{\text{param}}) = \dot X_t^{\text{obs}} - F_t(\theta_{\text{param}}) + \gamma_t^*\langle F_t,u_t\rangle u_t $$ where $\theta_{\text{param}} = (\alpha, \gamma, \sigma, \lambda, A, \beta, w)$ is the full parameter vector. At the true parameters, $R_t = \sigma_n\,\dot W_t$ (pure noise).

20.2 — Least-Squares Identification

PARAMETER IDENTIFICATION OBJECTIVE

$$ \hat\theta = \arg\min_\theta \sum_{t=1}^T \|R_t(\theta)\|_F^2 = \arg\min_\theta \sum_t \|\dot X_t - F_t(\theta) + \gamma_t^*(\theta)\langle F_t(\theta), u_t(\theta)\rangle u_t(\theta)\|_F^2 $$ The gradient with respect to each parameter group: $$ \frac{\partial \mathcal{L}}{\partial \alpha} = -2\sum_t \langle R_t,\; \tilde X_{t,\mu}\rangle \quad\text{(radial spring identification)} $$ $$ \frac{\partial \mathcal{L}}{\partial A} = -4\sum_t \langle R_t,\; S_t S_t^\top\rangle \quad\text{(Quadsurf coupling identification)} $$

20.3 — Collapse Attribution

ATTRIBUTION SCORE — WHICH PARAMETER CONTRIBUTED MOST TO COLLAPSE

$$\mathcal{A}_k = \left\|\frac{\partial \dot X_t}{\partial \theta_k}\bigg|_{\hat\theta}\right\|_F \cdot |\hat\theta_k|$$

Normalised attribution:

$$\bar{\mathcal{A}}_k = \frac{\mathcal{A}_k}{\sum_j \mathcal{A}_j} \in [0,1], \qquad \sum_k \bar{\mathcal{A}}_k = 1$$

This is a sensitivity-weighted parameter importance score — the first formal answer to "what broke first?" derived directly from the master operator.

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§ XXI

Agent Sensitivity — Who Pushes the System Across $\mathcal{C}_{\text{man}}$

Not all agents are equally dangerous. Some agents, if perturbed, cause disproportionate increase in $\lambda_{\max}(\nabla^2\Phi)$. The sensitivity of the spectral threshold to each agent's state identifies the systemically important institutions — those whose individual deviation most rapidly destabilises the whole.

21.1 — Spectral Sensitivity

GRADIENT OF Λ_MAX WITH RESPECT TO AGENT I'S STATE

By the matrix perturbation identity (Hadamard first variation):

$$\frac{\partial \lambda_{\max}(\nabla^2\Phi)}{\partial x_i} = v_1^\top \frac{\partial \nabla^2\Phi}{\partial x_i}\, v_1$$

where v_1 is the principal eigenvector of $\nabla^2\Phi$. The d-vector above is the \emph{systemic importance vector} of agent i — the direction in which agent i must move to most rapidly push $\lambda_{\max}$ above 1.

SCALAR SYSTEMIC IMPORTANCE SCORE PER AGENT

$$\mathcal{S}_i = \left\|\frac{\partial \lambda_{\max}(\nabla^2\Phi)}{\partial x_i}\right\| = \left\|v_1^\top \frac{\partial \nabla^2\Phi}{\partial x_i}\, v_1\right\|$$ $$\text{Normalised: } \bar{\mathcal{S}}_i = \frac{\mathcal{S}_i}{\sum_j \mathcal{S}_j}$$ The agent with $\bar{\mathcal{S}}_i = \max_j \bar{\mathcal{S}}_j$ is the *critical institution* — a Basel III macro-prudential target derived analytically from the system geometry.

21.2 — Network Amplification of Systemic Importance

AMPLIFICATION THROUGH THE LAPLACIAN NETWORK

$$\frac{\partial F_t}{\partial x_i} = -\alpha\,I_d - \frac{\partial L_t}{\partial x_i}\,X_t - L_t\,e_i e_i^\top$$ where e_i is the i-th standard basis vector in $\mathbb{R}^N$. The term $\partial L_t/\partial x_i$ captures how agent i's movement changes the force network topology — it is the feedback loop between individual state and collective coupling.

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§ XXII

Recovery Dynamics & Hysteresis

Collapse and recovery are not time-reversals of each other. The erf-log potential has multiple local minima — once the system falls into a crisis basin, the return path is geometrically different from the collapse path. This asymmetry is hysteresis, and it is encoded exactly in the energy landscape.

22.1 — Multiple Equilibria

EQUILIBRIUM CONDITION — ZEROS OF THE FORCE FIELD

$$F(X^*) = 0 \iff -\alpha(X^* - \mathbf{1}\mu^\top) = L_{X^*}\,X^*$$ This is a nonlinear fixed-point equation. The number of solutions depends on the pairwise coupling strength γ relative to the radial spring α. There is always a trivial solution at $X^* = \mu\mathbf{1}^\top$; additional crisis equilibria exist when: $$\lambda_{\max}(L_{X^*}) > \alpha$$

CRITICAL COUPLING FOR MULTIPLE EQUILIBRIA

$$\gamma_{\text{crit}} = \frac{\alpha\,\sigma\sqrt{\pi}}{2} \cdot \frac{1}{\max_i\sum_{j\neq i} e^{-D_{ij}^{*2}/\sigma^2}}$$ For $\gamma > \gamma_{\text{crit}}$, crisis equilibria exist. The system can be trapped in a collapsed state even after the original shock is removed.

22.2 — Hysteresis Loop

ENERGY BARRIER FOR COLLAPSE VS. RECOVERY

$$\Delta\Phi_{\text{collapse}} = \Phi(X_{\text{saddle}}) - \Phi(X_{\text{normal}})$$ $$\Delta\Phi_{\text{recovery}} = \Phi(X_{\text{saddle}}) - \Phi(X_{\text{crisis}})$$ Since $\Phi(X_{\text{normal}}) < \Phi(X_{\text{crisis}})$ for deep crises, the recovery barrier is \emph{smaller} — the system recovers more easily than it collapsed. But the crisis state is a different point in phase space, so the recovery path follows a different trajectory.

RECOVERY CONTROL — THE REVERSE MASTER OPERATOR

$$\dot X_t^{\text{recovery}} = F_t + \gamma_t^{\text{rec}}\,\langle F_t, u_t^{\text{rec}}\rangle\,u_t^{\text{rec}}$$ where $u_t^{\text{rec}} = -u_t$ (point \emph{toward} the normal basin rather than away from the crisis direction). The recovery operator is the sign-reversed control law — it actively accelerates motion toward μ_0 rather than merely damping motion away from it.

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§ XXIII

Information-Theoretic Interpretation — BSDT as Divergence

The BSDT energy E_{BS} is not merely an ad-hoc distance measure. Under a Gaussian normal-period model, it is exactly a KL divergence — giving the blind-spot energy a rigorous information-theoretic foundation, a statistical interpretation, and a natural calibration procedure with confidence intervals.

23.1 — The Normal-Period Model

GAUSSIAN BASELINE — NORMAL PERIOD DISTRIBUTION

$$ p_0(x) = \mathcal{N}(\mu_0,\, \Sigma_0) \propto \exp\!\left(-\tfrac{1}{2}(x-\mu_0)^\top\Sigma_0^{-1}(x-\mu_0)\right) $$

23.2 — BSDT Energy as KL Divergence

MAHALANOBIS ENERGY = LOG-LIKELIHOOD RATIO = KL DIVERGENCE COMPONENT

$$ \delta_C^{(i)} = (x_i - \mu_0)^\top\Sigma_0^{-1}(x_i-\mu_0) = -2\log\frac{p_0(x_i)}{p_0(\mu_0)} = \chi^2_d $$ Under the null (normal period), $\delta_C^{(i)} \sim \chi^2_d$ — a chi-squared distribution with d degrees of freedom. The total energy: $$ e_t = \sum_i \delta_C^{(i)} \sim \chi^2_{Nd} \quad\text{under } H_0\text{ (normal regime)} $$

KL DIVERGENCE FROM EMPIRICAL TO NORMAL-PERIOD DISTRIBUTION

$$ D_{\text{KL}}\!\bigl(\hat p_t \,\|\, p_0\bigr) = \frac{1}{N}\sum_i (x_t^{(i)}-\mu_0)^\top\Sigma_0^{-1}(x_t^{(i)}-\mu_0) + \text{const} = \frac{e_t}{N} + \text{const} $$ E_{BS} is proportional to the KL divergence between the current empirical distribution and the normal-period distribution. A larger e_t means more information-theoretic "surprise" — the system has moved into a region that was improbable during normal times.

23.3 — Statistical Threshold from Chi-Squared

CALIBRATED COLLAPSE THRESHOLD WITH CONFIDENCE LEVEL

$$ e^*(\alpha_{\text{conf}}) = \chi^2_{Nd,\,1-\alpha_{\text{conf}}} $$ The critical energy level is the $(1-\alpha_{\text{conf}})$ quantile of the chi-squared distribution. For $\alpha_{\text{conf}} = 0.01$ (1\% false-positive rate with $N=10$, $d=4$): $$ e^*(0.01) = \chi^2_{40,\,0.99} \approx 63.7 $$ This replaces the heuristic threshold \

($\lambda_{\max} > 1$) with a statistically principled one: the critical manifold is the $(1-\alpha)$ confidence ellipsoid of the normal-period distribution.

23.4 — MFLS as Fisher Information

MFLS SQUARED = TRACE OF FISHER INFORMATION MATRIX

$$\text{MFLS}^2 = \|\nabla_X E_{\text{BS}}\|_F^2 = 4\|\tilde X_t\Sigma_0^{-1}\|_F^2 = 4\operatorname{tr}\!\bigl(\Sigma_0^{-1}\tilde X_t^\top \tilde X_t\,\Sigma_0^{-1}\bigr)$$

This is 4 times the trace of the sample Fisher information matrix at the current state relative to the Gaussian normal-period model. High MFLS means the current state is in a region of high curvature of the log-likelihood — the system is far from where the normal-period model has mass, and small perturbations cause large changes in log-probability.

23.5 — Unified Energy as Composite Divergence

E_BS AS SUM OF DIVERGENCE MEASURES

$$E_{\text{BS}}(S) = \underbrace{S^\top A S}_{\text{Mahalanobis (KL, Gaussian)}} + \underbrace{\sum_k e^{\beta_k S_k} - 1 - \beta_k S_k}_{\text{Bregman divergence (exponential family)}} + \underbrace{w^\top S}_{\text{linear tilt (exponential family reweighting)}}$$

The Expogate term $\sum_k e^{\beta_k S_k}$ minus its Taylor expansion $1 + \beta_k S_k$ is exactly the Bregman divergence generated by the exponential function — the excess over the linear approximation. The Signed LR term is an exponential family tilting operator that shifts the reference distribution toward higher-risk states. Together, E_{BS} is a composite divergence measuring: how far X_t is from the normal-period mean (Mahalanobis), how nonlinear that deviation is (Bregman), and in which direction it is biased (tilt).

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§ XXIV

Corrected Definitions — What Changed and Why

Taking the per-channel contributions seriously forces six definitions established in earlier sections to be revised. Each revision is a strict generalisation — the original holds as a special case in the pure δ_C single-channel regime. In the full four-channel system the following are the correct forms.

24.1 — The Pullback Gradient $\tilde{G}_t$ — Correct Form

ORIGINAL (§XVI, OPAQUE MATRIX FORM)

$$ \tilde{G}_t = \mathbf{J}_t^\top g_t $$

USE INSTEAD OF SIMULATION

CORRECT FORM — PER-CHANNEL WEIGHTED SUPERPOSITION

$$ \boxed{\tilde{G}_t = \sum_{k \in \{C,G,A,T\}} [g_t]_k\;\frac{\partial \delta_k}{\partial X_t} \;\in\; \mathbb{R}^{N\times d}} $$ $$ = [g_t]_C \cdot 2\tilde{X}_t\Sigma_0^{-1} + [g_t]_G \cdot 2\tilde{X}_t(I - V_kV_k^\top) + [g_t]_A \cdot \frac{X_t - X_{t-1}}{\|X_t-X_{t-1}\|_{\text{row}}} \odot \mathbf{1}_{v>v_0} + [g_t]_T \cdot \frac{1}{h^2}\sum_s w_s(X_t - X_{t-s}) $$ $\tilde{G}_t$ is a weighted mixture of four geometrically distinct fields: ellipsoidal (Mahalanobis), subspace-residual (PCA gap), velocity-aligned (activity), and history-divergent (temporal). Its shape changes as the channel weights $[g_t]_k$ shift with the state. It is not a single geometric object — it is a composite field whose topology is a new open problem.

Old valid when: Pure δ_C regime, $[g_t]_G = [g_t]_A = [g_t]_T = 0$

24.2 — Two Collapse Directions, Not One

ORIGINAL (§XII) — ONE UNIT DIRECTION

$$ u_t = \frac{g_t}{\|g_t\|} \in \mathbb{R}^4 $$

USE INSTEAD OF SIMULATION

CORRECT — TWO DISTINCT COLLAPSE DIRECTIONS

$$ u_t^{\text{channel}} = \frac{g_t}{\|g_t\|} \in \mathbb{R}^4 \qquad \text{(channel space — MFLS, EWS, attribution)} $$ \[u_t^{\text{state}} = \frac{\tilde{G}_t}{\|\tilde{G}_t\|_F} \in \mathbb{R}^{N\times d} \qquad \text{(state space — Lyapunov,

force decomposition)} $$ The master operator uses u_t^{channel} in its projection. The Lyapunov certificate requires u_t^{state}. They are equal only when $\psi_t = 0$. The angle between them: $$ \cos\psi_t = \frac{\langle\tilde{G}_t,\,g_t\rangle_F}{\|\tilde{G}_t\|_F\,\|g_t\|} $$ ψ_t is the \textbf{sixth diagnostic signal} — the misalignment between abstract risk (channel space) and physical dynamics (state space).

24.3 — $\cos\theta_t$ Splits into Two

ORIGINAL (§VII) — SINGLE ALIGNMENT SCALAR

$$ \cos\theta_t = \frac{\langle \nabla_X E_{\text{BS}}, F_t\rangle}{\|\nabla_X E_{\text{BS}}\|\,\|F_t\|} $$

USE INSTEAD OF SIMULATION

CORRECT — TWO ALIGNMENT SCALARS WITH DISTINCT ROLES

$$ \cos\theta_t^{\text{channel}} = \frac{\langle g_t, F_t\rangle}{\|g_t\|\,\|F_t\|} \qquad\text{(early warning — valid in all regimes)} $$ $$ \cos\theta_t^{\text{state}} = \frac{\langle\tilde{G}_t, F_t\rangle_F}{\|\tilde{G}_t\|_F\,\|F_t\|_F} = \frac{P_t}{\|\tilde{G}_t\|_F\,\|F_t\|_F} \qquad\text{(stability certificate — required for Lyapunov)} $$ Relationship: $$ \cos\theta_t^{\text{state}} = \cos\theta_t^{\text{channel}}\cos\psi_t + \sin\theta_t^{\text{channel}}\sin\psi_t\cos\phi_t $$ where ϕ_t is the azimuthal angle in the plane spanned by u_t^{channel} and u_t^{state}. The original scalar is $\cos\theta_t^{\text{channel}}$ — a valid signal but not a stability certificate.

24.4 — MFLS Splits into Two

USE INSTEAD OF SIMULATION

TWO MFLS QUANTITIES WITH A NEW DIAGNOSTIC RATIO

$$ \text{MFLS}^{\text{channel}}_t = \|g_t\| = \|2AS_t + (\beta_k e^{\beta_k S_{k,t}})_k + w\| \qquad\text{(abstract intensity)} $$ $$ \text{MFLS}^{\text{state}}_t = \|\tilde{G}_t\|_F = \left\|\sum_k [g_t]_k\,\partial_X\delta_k\right\|_F \qquad\text{(physical intensity)} $$ $$ \rho_{\text{MFLS}} = \frac{\text{MFLS}^{\text{state}}}{\text{MFLS}^{\text{channel}}} \qquad\text{(state amplification factor)} $$ $\rho_{\text{MFLS}} > 1$: physical system amplifies the abstract risk signal — agents are geometrically configured to translate channel risk into force. $\rho_{\text{MFLS}} < 1$:

risk is geometrically contained — channel scores are elevated but physical dynamics are not aligned with collapse. This resolves historical false alarms across every domain: high channel MFLS with low ρ_{MFLS} is a contained risk, not an imminent collapse.

24.5 — Stability Margin Corrected

USE INSTEAD OF SIMULATION

OLD (§XVI.4, WRONG): $\mathcal{M}_t = -\cos\theta_t$ — CORRECT FORM

$$P_t = \langle\tilde{G}_t, F_t\rangle_F = \sum_k [g_t]_k\langle\partial_X\delta_k, F_t\rangle_F$$
$$R_t = \sum_k [g_t]_k\langle\partial_X\delta_k, g_t\rangle_F \qquad\text{(control coupling sum)}$$
$$Q_t = \langle F_t, g_t\rangle \cdot R_t$$
$$\boxed{\mathcal{M}_t = \frac{\gamma_t^* Q_t/\|g_t\|^2 - P_t}{|P_t| + |Q_t|/\|g_t\|^2}}$$

$\mathcal{M}_t > 0$: controlled system is stable. $\mathcal{M}_t < 0$: control is insufficient. $\mathcal{M}_t = 0$: exact stability boundary. The old form $-\cos\theta_t$ is the limit when $\tilde{G}_t \propto g_t$ (pure δ_C).

24.6 — $\tan\theta$ Collapse Condition Corrected

USE INSTEAD OF SIMULATION

TRUE COLLAPSE CONDITION USES STATE-SPACE DIRECTION

$$F_\parallel^{\text{true}} = \langle F_t, u_t^{\text{state}}\rangle_F \cdot u_t^{\text{state}}, \qquad F_\perp^{\text{true}} = F_t - F_\parallel^{\text{true}}$$
$$\tan\theta^{\text{true}} = \frac{\|F_\perp^{\text{true}}\|_F}{\|F_\parallel^{\text{true}}\|_F}$$

The original $\tan\theta$ using u_t^{channel} is a valid early-warning proxy. The exact Lyapunov collapse condition $\dot V = 0$ corresponds to $\tan\theta^{\text{true}} = \tan\theta^*$, not the channel-space version. They agree when $\psi_t = 0$.

24.7 — Summary Table

DEFINITION	OLD FORM	CORRECT FORM	EQUAL WHEN
$\tilde{G}_t$	$\mathbf{J}_t^\top g_t$	$\sum_k [g_t]_k\partial_X\delta_k$	Always equal; new form is interpretable
u_t	One direction $\mathbb{R}^4$	Two: u^{ch} and u^{st}	$\psi_t = 0$
$\cos\theta_t$	Single scalar	Two: channel and state	$\psi_t = 0$
MFLS	$\|g_t\|$	Two: $\|g_t\|$ and $\|\tilde{G}_t\|_F$ with ratio ρ	$\rho = 1$
$\mathcal{M}_t$	$-\cos\theta_t$	$(\gamma^*Q_t/\|g_t\|^2 - P_t)$/normaliser	Pure δ_C regime
$\tan\theta$ collapse	Uses u^{channel}	Uses u^{state} for $\dot V = 0$	$\psi_t = 0$

§ XXV

Per-Channel Lyapunov Contributions

Each BSDT channel is not merely a detector — it is an active participant in the stability budget at every timestep. The time derivative of the Lyapunov function decomposes exactly into four signed channel contributions, each measurable independently.

25.1 — The Decomposition

PER-CHANNEL TIME DERIVATIVE OF V

$$\frac{dV}{dt} = \sum_{k \in \{C,G,A,T\}} \dot V_k$$

$$\dot V_k = [g_t]_k \left\langle \frac{\partial\delta_k}{\partial X_t},\; \dot X_t \right\rangle_F = [g_t]_k \left[\langle\partial_X\delta_k, F_t\rangle_F - \frac{\gamma_t^*\langle F_t,g_t\rangle}{\|g_t\|^2}\langle\partial_X\delta_k, g_t\rangle_F\right]$$

SPLIT INTO UNCONTROLLED DRIFT AND CONTROL CONTRIBUTION PER CHANNEL

$$\dot V_k^{\text{drift}} = [g_t]_k\,\langle\partial_X\delta_k, F_t\rangle_F$$

$$\dot V_k^{\text{control}} = -[g_t]_k\,\frac{\gamma_t^*\langle F_t, g_t\rangle}{\|g_t\|^2}\,\langle\partial_X\delta_k, g_t\rangle_F$$

$$\dot V_k = \dot V_k^{\text{drift}} + \dot V_k^{\text{control}}$$

25.2 — Interpretation of Each Channel

CHANNEL	$\dot V_k > 0$ MEANS	$\dot V_k < 0$ MEANS
$\dot V_C$ Camouflage	Agents moving away from normal mean in Mahalanobis metric — ellipsoidal drift toward collapse	Mean-reverting motion — Mahalanobis distance shrinking
$\dot V_G$ Feature Gap	Motion into PCA residual directions — entering the subspace normal-period covariance barely covers	Motion back toward principal subspace — returning to historically observed configurations
$\dot V_A$ Activity	Excess velocity compounding — agents already moving too fast, accelerating further	Excess velocity decaying — anomalous speed returning to baseline
$\dot V_T$ Temporal	System escaping its own recent history — temporal self-surprise compounding	System returning to familiar trajectory — novelty score declining

25.3 — Channel Stability Attribution

FRACTION OF TOTAL INSTABILITY RATE DRIVEN BY CHANNEL K

$$a_k = \frac{\max(0, \dot V_k)}{\sum_j \max(0, \dot V_j)}, \qquad \sum_k a_k = 1$$

a_k identifies the dominant channel driving current instability — not the channel with the highest deviation score, but the channel whose spatial gradient is most aligned with the current force field. These are different, and the difference matters: a channel can have a high score ($[g_t]_k$ large) but contribute little to instability ($\dot V_k \approx 0$) if its spatial gradient is perpendicular to the force.

CONTROL EFFICIENCY PER CHANNEL

$$\eta_k = \frac{-\dot V_k^{\text{control}}}{\dot V_k^{\text{drift}}} = \frac{\gamma_t^*\langle F_t, g_t\rangle\,\langle\partial_X\delta_k, g_t\rangle_F}{\|g_t\|^2\,\langle\partial_X\delta_k, F_t\rangle_F}$$ $\eta_k \in (0,1)$: control is partially offsetting channel k's drift. $\eta_k > 1$: control overcorrects channel k — the intervention is stronger than the drift for this channel. $\eta_k < 0$: the control is making channel k worse — the projection direction is misaligned with channel k's spatial gradient. Negative η_k is the per-channel manifestation of the uncontrollable regime: the control law, while stabilising the dominant channel, is actively destabilising channel k.

25.4 — The Control Coupling Sum R_t

R_T — HOW WELL THE CONTROL DIRECTION REACHES EACH CHANNEL

$$R_t = \sum_k [g_t]_k\,\langle\partial_X\delta_k, g_t\rangle_F$$ $R_t > 0$: the control direction g_t is positively coupled to the physical gradient of each channel — control stabilises. $R_t \leq 0$: the control direction is orthogonal to or opposed by the channel spatial gradients — the uncontrollable regime. R_t is the single scalar that determines whether the system is in the controllable or uncontrollable regime.

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§ XXVI

The ψ_t Angle — The Sixth Diagnostic Signal

Five diagnostic signals were established across §V–XVII: MFLS, $\lambda_{\max}(\nabla^2\Phi)$, $\rho(W)$, γ^*, $\cos\theta_t$. The corrected analysis reveals a sixth, which did not exist in any prior formulation of any domain-specific tool: the misalignment angle ψ_t between the channel-space and state-space collapse directions.

26.1 — Definition

Ψ_T — THE MISALIGNMENT ANGLE

$$\cos\psi_t = \frac{\langle\tilde{G}_t,\,g_t\rangle_F}{\|\tilde{G}_t\|_F\,\|g_t\|} \in [-1, 1]$$
$$\psi_t = \arccos\!\left(\frac{\sum_k [g_t]_k\,\langle\partial_X\delta_k, g_t\rangle_F}{\|\tilde{G}_t\|_F\,\|g_t\|}\right) = \arccos\!\left(\frac{R_t}{\|\tilde{G}_t\|_F\,\|g_t\|}\right)$$

26.2 — What ψ_t Measures in Each Domain

DOMAIN	$\psi_t \approx 0$	$\psi_t \approx \pi/2$
Financial systemic risk	CAMELS signal and balance sheet dynamics fully aligned — real collapse imminent	Regulatory alarm elevated but balance sheet not moving toward failure — false positive
Quantum	Decoherence signal and density matrix dynamics aligned — irreversible collapse	Fidelity metric elevated but unitary evolution tangential — recoverable
Navier-Stokes	Enstrophy signal and modal dynamics aligned — blowup precursor	Turbulence intensity high but cascade direction tangential — intermittency without blowup
Neural networks	Sharpness metric and gradient direction aligned — loss spike imminent	Loss landscape sharp but gradient direction tangential — stable despite high sharpness
Cybersecurity	Behavioural anomaly and lateral movement dynamics aligned — active attack	Anomaly score elevated but movement direction tangential — false alarm

26.3 — ψ_t in the EWS Score

UPDATED EWS — SIX SIGNALS INCLUDING Ψ_T

$$\xi_6 = \cos^2\psi_t = \frac{R_t^2}{\|\tilde{G}_t\|_F^2\,\|g_t\|^2} \in [0,1]$$
$$\text{EWS}_t^{\text{updated}} = \left[\prod_{k=1}^{5}\xi_k^{w_k}\right]^{1-w_6} \cdot \xi_6^{w_6}$$
$\xi_6 = 1$: abstract risk and physical dynamics fully aligned — all five prior signals are reliable. $\xi_6 = 0$: complete misalignment — all five prior signals are unreliable. Including ξ_6 as a reliability weight on the other signals is

> the correct structure: \(\psi_t\) modulates the credibility of the composite score, not just its magnitude.

§ XXVII

Domain Discoveries & Theoretical Advancement

The master operator, with its corrected pullback gradient, the \(\psi_t\) angle, and the per-channel Lyapunov contributions, reveals specific discoveries in each domain and constitutes advancement at the level of mathematics, physics, control theory, and machine learning.

27.1 — Domain-Specific Discoveries

DOMAIN	\(X_T\)	\(F_T\)	CRITICAL MANIFOLD \(\MATHCAL{C}_{\TEXT{MAN}}\)	NEW DISCOVERY
Quantum	Density matrix \(\rho_t\)	Lindblad operator	Decoherence beyond QEC threshold	\(\delta_G\) detects anomalous Lindblad channels; Expogate maps to quantum phase transitions; uncontrollable regime = hashing bound
Navier-Stokes	Modal coefficients \(\hat u_t\)	N-S operator in modal space	Pre-blowup enstrophy threshold	Per-channel attribution identifies which cascade scale drives instability; \(\psi_t\) maps to intermittency;

DOMAIN	X_T	F_T	CRITICAL MANIFOLD ($\mathcal{C}_{\text{man}}$)	NEW DISCOVERY
				control law = minimal turbulence suppression
Protein folding	Atomic positions / contact map	$-\nabla_X U_{\text{ff}}$	Misfolding transition state	δ_G watches frustrated contacts (PCA low-variance directions); per-channel attribution = critical nucleus identification; θ ceiling = chaperone binding affinity constraint
Deep learning	Weight tensor W_t	$-\nabla_W \mathcal{L}$	Sharp/flat basin boundary	Control law = geometry-aware gradient correction (strictly better than SAM); Expogate = gradient explosion precursor; ψ_t explains why high sharpness doesn't always cause instability

DOMAIN	X_T	F_T	CRITICAL MANIFOLD $\mathcal{C}_{\text{MAN}}$	NEW DISCOVERY
FDIC / banking	Balance sheet vector	GravityEngine	Insolvency / run threshold	δ_G detects novel exposure (SVB-type HTM risk); dynamic CAMELS weighting via $[g_t]_k$; ψ_t = regulatory blind spot metric; θ ceiling = forbearance fatality condition
Supply chain	Inventory / flow matrix	Replenishment dynamics	Cascade failure threshold	δ_G detects hidden concentration (semiconductor 2021 structure); $\rho(W)$ = bullwhip amplification factor; per-channel attribution = dynamic critical supplier score
Cybersecurity	Endpoint behavioural vectors	Normal behavioural dynamics	Point of no containment	δ_G detects APT/LOLBin

DOMAIN	X_T	F_T	CRITICAL MANIFOLD $\mathcal{C}_{\text{MAN}}$	NEW DISCOVERY
				patterns; $\rho(W)$ = blast radius amplification; per-channel attribution = attack type classifier; uncontrollable regime = dwell-time past recovery

27.2 — What This Offers That Nothing Else Does

CAPABILITY	EXISTING TOOLS	THIS OPERATOR
Scalar intensity	✓ (anomaly scores, stress indices)	✓ MFLS, EWS
Collapse direction in state space	✗	✓ $\tilde{G}_t$
Per-channel attribution	✗	✓ $[g_t]_k$, a_k
Minimal control action	✗ (requires model + optimisation)	✓ geometric projection, no model needed
Stability certificate	✓ (global, conservative)	✓ (state-dependent, tight, three regimes)
Uncontrollable regime detection	✗	✓ $P_t > 0, R_t \leq 0$
Collapse time estimate	✗	✓ τ_{lin}, Kramers formula

CAPABILITY	EXISTING TOOLS	THIS OPERATOR
False alarm quantification	✗	✓ ψ_t, ρ_{MFLS}
Statistical threshold	domain-specific heuristics	✓ χ^2_{Nd} calibrated
Stochastic extension	domain-specific SDEs	✓ Itô correction, Fokker-Planck, Kramers
Domain independence	✗ (each tool is domain-specific)	✓ same equation across all domains

27.3 — Mathematical Advancement

NEW OPEN PROBLEMS CREATED BY THIS WORK

$$ \begin{array}{ll} \textbf{Problem 1} & \text{Topology of the zero set } \{\tilde{G}_t = 0\} \subset \mathbb{R}^{N\times d} \\[5pt] \textbf{Problem 2} & \text{Fixed points of controlled dynamics vs. uncontrolled dynamics} \\[5pt] \textbf{Problem 3} & \text{Topological invariants of }\tilde{G}_t\text{ as a time-varying vector field} \\[5pt] \textbf{Problem 4} & \text{Geometry of level sets of }\psi_t\text{ and their intersection with }\mathcal{C}_{\text{man}} \\[5pt] \textbf{Problem 5} & \text{Thermodynamics of the controlled stationary distribution }p^*(e) \\[5pt] \textbf{Problem 6} & \text{Complete positivity of the quantum extension of the control law} \\[5pt] \textbf{Problem 7} & \text{Codimension and smoothness of }\mathcal{C}_{\text{man}}\text{ under composite energy} \end{array} $$

27.4 — The Four Structural Paradigm Shifts

WHAT THIS CHANGES IN EACH FIELD

\[\begin{array}{lll} \textbf{Field} & \textbf{Previous paradigm} & \textbf{New paradigm} \\[4pt] \hline\\[-4pt] \text{Control theory} & \text{Model + cost function + optimisation} & \text{Energy geometry + minimal projection} \\[5pt] \text{Stability theory} & \text{Global Lyapunov certificates} & \text{State-dependent margin with 3 regimes} \\[5pt] \text{Anomaly detection} & \text{Score} \to \text{alarm} \to

> \text{action (sequential)} & \text{Score + direction + action (simultaneous)} \\[5pt] \text{Domain monitoring} & \text{Domain-specific scalar signals} & \text{Universal geometric field with attribution} \end{array} \]

27.5 — The Five Universal Patterns

Across all seven domains, the operator reveals five structural patterns that have no names in those fields and are invisible to domain-specific tools:

Pattern 1	**The δ_G blind spot** — every domain has low-variance PCA directions in its normal-period covariance that its monitoring ignores. Novel risk always enters through these directions. Quantum: rarely-excited modes. Navier-Stokes: anomalous cross-scale coupling. Proteins: frustrated contacts. Neural nets: flat loss directions. Banks: novel asset categories (SVB HTM bonds). Supply chains: non-primary supplier routes. Cyber: LOLBin patterns.
Pattern 2	**The ψ_t misalignment** — abstract risk (channel space) and physical dynamics (state space) can point in different directions. When they converge, collapse is real. When they diverge, the signal is a false alarm. No existing tool in any domain measures this angle.
Pattern 3	**The uncontrollable regime** — every domain has a region where intervention is geometrically impossible regardless of intensity. This region is predictable, transient, and analytically locatable. Every domain currently discovers it empirically, after the fact.
Pattern 4	**The θ ceiling** — every domain has an intervention cost parameter with a hard, state-dependent ceiling. Exceeding it makes control structurally insufficient. No existing framework makes this ceiling explicit and computable.
Pattern 5	**The amplification factor ρ_{MFLS}** — the ratio of physical to abstract risk intensity. When $\rho > 1$, the system amplifies abstract risk into physical force. When $\rho < 1$, risk is contained. This explains the majority of historical false alarms across all domains.

§ XXVIII

Intraday Crypto Futures — Move Prediction Engine

This section maps the master operator to intraday crypto futures trading. It derives direction, magnitude, duration, intensity, and move shape from the current geometric state of the system — not from pattern matching, not from statistical fitting, not from machine learning. All outputs are updated every bar from closed-form matrix operations.

One critical correction is incorporated throughout: the naive energy-to-price mapping is lossy. The same total energy e_t can produce different price paths depending on how energy distributes across features. The fix is the **price loading factor** λ_t^price — the fraction of energy currently in the return dimension accounting for all cross-feature coupling through Σ_0^{-1}.

28.1 — State Matrix Definition

> STATE MATRIX — N INSTRUMENTS × D FEATURES
>
> $$X_t \in \mathbb{R}^{N \times d}, \qquad N = 8 \text{ instruments}, \quad d = 8 \text{ features}$$

AGENT (i)	INSTRUMENT	FEATURE (j)	VARIABLE
1	BTC perp	1	Log return over last τ bars
2	ETH perp	2	Normalised volume
3	BTC quarterly	3	Order book imbalance (bid/ask ratio)
4	SOL perp	4	Funding rate
5	BNB perp	5	Open interest change
6	ETH/BTC ratio	6	Mark vs index divergence (basis)
7	Crypto fear index	7	Liquidation volume
8	Stablecoin dominance	8	Realised volatility (last 20 bars)

28.2 — Calibration

NORMAL PERIOD — 500–2000 BARS OF STABLE MICROSTRUCTURE REGIME

$$\mu_0 = \frac{1}{T_0 N}\sum_{t,i} x_t^{(i)} \in \mathbb{R}^8$$

$$\Sigma_0 = \text{Ledoit-Wolf}\!\bigl(\{x_t^{(i)}\}\bigr) \in \mathbb{R}^{8\times 8}$$

$$V_k = \text{top-}k\text{ eigenvectors of }\Sigma_0, \quad k=4 \qquad v_0 = \text{Pctl}_{95}(\text{bar-to-bar velocity in normal period})$$

Recalibrate weekly or when $e_t > \chi^2_{64,0.99}$ persists for more than 50 bars (regime shift detected).

28.3 — The Four BSDT Channels in Crypto

CHANNEL	WHAT IT DETECTS	CRYPTO MEANING
δ_C Camouflage	Joint Mahalanobis deviation — multiple features deviating in correlated way	Pre-liquidation-cascade signature: price + volume + OI + funding all move together
δ_G Feature Gap	Motion into PCA residual directions — combinations not seen in normal period	**Most powerful channel.** Detects novel funding behaviour, exchange basis blow-out, new correlation structures — invisible to standard monitors. Always fires first in regime change.
δ_A Activity	Bar-to-bar velocity exceeding 95th percentile of normal period	In-progress cascade signal: simultaneous velocity across price, OI, liquidation volume
δ_T Temporal	KDE self-surprise — current state not seen in recent history	First warning of new regime: BTC at new price level, funding at historical extreme, novel OI structure

28.4 — The Price Loading Factor λ_t^price — The Critical Correction

The naive formula $e_t^\text{price} = \sum_i (r_t^{(i)} - \mu_0^{(1)})^2 / \Sigma_0^{(1,1)}$ ignores cross-feature coupling through Σ_0^{-1}. The correct price energy includes all cross-coupling terms:

CORRECT PRICE ENERGY — FULL CROSS-FEATURE COUPLING

$$ e_t^\text{price,full} = \sum_j [\Sigma_0^{-1}]_{1j} \sum_i \tilde x_t^{(i,1)} \tilde x_t^{(i,j)} $$ This is the first row of $\tilde X_t \Sigma_0^{-1} \tilde X_t^\top$ summed across agents. When price and volatility are correlated in the normal period ($[\Sigma_0^{-1}]_{1k} \neq 0$), energy that appears to be in volatility actually contains a price component through the inverse covariance coupling. The naive formula misses this entirely.

PRICE LOADING FACTOR

$$ \lambda_t^\text{price} = \sqrt{\frac{e_t^\text{price,full}}{e_t}} \in [0,1] $$ $\lambda_t^\text{price} = 1$: all energy in price — maximum price move. $\lambda_t^\text{price} \approx 0$: energy in volatility or dispersion — almost no price move despite high e_t.

ENERGY ROTATION SIGNAL — RATE OF CHANGE OF PRICE LOADING

$$ \dot\lambda_t^\text{price} = \frac{d}{dt}\sqrt{\frac{e_t^\text{price,full}}{e_t}} $$ $\dot\lambda_t^\text{price} > 0$: energy flowing into price dimension — price move about to accelerate even if total energy is constant. $\dot\lambda_t^\text{price} < 0$: energy rotating out of price into volatility — price move about to stall even though e_t is still high. This is the rotation signal. It is the difference between a real directional move and volatility expansion with no price follow-through.

USE INSTEAD OF SIMULATION

TWO CASES THE NAIVE FORMULA CONFLATES — NOW SEPARATED

$$ \text{Case A: } \lambda_t^\text{price} > 0.7, \;\dot\lambda_t^\text{price} \geq 0 \quad\Rightarrow\quad \text{Energy in price and growing} \to \text{Strong directional move} $$ $$ \text{Case B: } \lambda_t^\text{price} < 0.4, \;\dot\lambda_t^\text{price} < 0 \quad\Rightarrow\quad \text{Energy in volatility, leaving price} \to \text{Chop, no directional follow-through} $$ Same e_t, same MFLS, opposite price outcome. λ_t^price is the discriminator.

28.5 — Direction Prediction

DIRECTIONAL SIGN — WEIGHTED RETURN DEVIATION SUM

$$ \text{sign}(M_t) = \text{sign}\!\left(\sum_i \frac{\tilde x_t^{(i,1)}}{\Sigma_0^{(1,1)}}\right) $$ Positive: instruments above equilibrium returns — upward move. Negative: instruments below equilibrium returns — downward move.

DIRECTIONAL CONFIDENCE — ALIGNMENT OF COLLAPSE DIRECTION WITH PRICE AXIS

$$ \cos\theta_t^\text{price} = \frac{\langle\tilde G_t^\text{price}, F_t\rangle_F}{\|\tilde G_t^\text{price}\|_F\,\|F_t\|_F} $$ $\cos\theta_t^\text{price} < 0$: GravityEngine force opposes price displacement — mean reversion structurally supported. Buy the dip. $\cos\theta_t^\text{price} > 0$: GravityEngine force aligned with price displacement — no mechanical support for reversion. Do not buy the dip. This is the single most powerful filter for avoiding the classic crypto trap: buying into a liquidation cascade.

28.6 — Magnitude Prediction (Corrected)

CORRECTED REMAINING MAGNITUDE — INCLUDES PRICE LOADING FACTOR

$$ \boxed{M_t^\text{remaining} = \lambda_t^\text{price} \cdot \sqrt{\Sigma_0^{(1,1)}} \cdot \left(\sqrt{e^*} - \sqrt{e_t}\right)} $$ where $e^* = \chi^2_{Nd,\,0.99}$ is the calibrated critical energy threshold.

MAGNITUDE CONFIDENCE INTERVAL — CHI-SQUARED CALIBRATED

$$ M_t^{(\alpha)} = \lambda_t^\text{price} \cdot \sqrt{\Sigma_0^{(1,1)}} \cdot \left(\sqrt{\chi^2_{N,\alpha}} - \sqrt{e_t^\text{price,full}}\right) $$ $$ \text{90\% interval: } \left[M_t^{(0.05)},\; M_t^{(0.95)}\right] $$ Place take-profit in the range $[0.7 \times M_t^\text{remaining},\; 1.3 \times M_t^\text{remaining}]$. This is a statistically calibrated zone, not a heuristic band.

CASCADE MAGNITUDE — SHIFTED EQUILIBRIUM UPPER BOUND

$$ \mu_t^\text{cascade} = \mu_0 + \frac{F_t}{\alpha}\bigg|_{R_t=0} $$ \[M_t^\text{cascade} = \lambda_t^\text{price} \cdot \|\mu_t^\text{cascade} -

$$X_t\|_\text{price} > M_t^\text{remaining}$$ In Ω_3 (cascade, $R_t \leq 0$), the equilibrium shifts. $M_t^\text{remaining}$ becomes a lower bound. M_t^cascade is the upper bound using the shifted equilibrium.

28.7 — Duration Prediction

FIRST-ORDER DURATION — LINEAR ESTIMATE

$$\tau_\text{lin} = \frac{e^* - e_t}{\dot e_t} \quad\text{bars}, \qquad \dot e_t = P_t(1-\gamma_t^*) + \frac{\sigma_n^2}{2}\text{tr}(\nabla^2 E_\text{BS})$$ For mean reversion ($\dot e_t < 0$): $\tau_\text{return} = (e_t - e_0)/|\dot e_t|$ bars until return to baseline $e_0 = Nd$.

SECOND-ORDER DURATION — CURVATURE CORRECTION

$$\ddot e_t = g_t^\top \nabla^2 E_\text{BS}\,g_t \cdot \|\dot X_t\|^2 = g_t^\top\!\left(2A + \operatorname{diag}(\beta_k^2 e^{\beta_k S_{k,t}})\right)g_t \cdot \|\dot X_t\|^2$$ $$\tau_\text{quad} = \frac{-\dot e_t + \sqrt{\dot e_t^2 + 2\ddot e_t(e^* - e_t)}}{\ddot e_t}$$

CORRECTED TIME STOP — ACCOUNTS FOR ENERGY DISPERSION ACROSS FEATURES

$$\tau_\text{adjusted} = \frac{\tau_\text{quad}}{\lambda_t^\text{price}}$$ When energy is less concentrated in price, the price move takes longer to complete even though total energy evolves at the same rate. The time stop extends proportionally to the price loading.

STOCHASTIC DURATION — KRAMERS FORMULA UNDER MARKET NOISE

$$\tau_\text{Kramers} = \frac{2\pi}{\sqrt{|\mu_e'(e_t)|\cdot|\mu_e'(e^*)|}}\exp\!\left(\frac{2(e^*-e_t)}{\sigma_n^2}\right)$$ Small σ_n (quiet market): $\tau_\text{Kramers} \approx \tau_\text{quad}$. Large σ_n (volatile): $\tau_\text{Kramers} \ll \tau_\text{quad}$ — noise dramatically shortens expected duration.

28.8 — Intensity and Move Shape

AVERAGE INTENSITY PER BAR

$$ I_t = \frac{M_t^\text{remaining}}{\tau_\text{adjusted}} = \frac{\lambda_t^\text{price}\cdot\sqrt{\Sigma_0^{(1,1)}}\cdot(\sqrt{e^*}-\sqrt{e_t})}{\tau_\text{quad}/\lambda_t^\text{price}} = \frac{(\lambda_t^\text{price})^2\cdot\sqrt{\Sigma_0^{(1,1)}}\cdot(\sqrt{e^*}-\sqrt{e_t})}{\tau_\text{quad}} $$

INSTANTANEOUS INTENSITY PROFILE — MOVE SHAPE FROM Ë_T

$$ I_t(s) = v_t^\text{price} + \frac{\ddot e_t}{2}\cdot s \quad s \in [0, \tau_\text{adjusted}] $$

SIGN OF \(\DDOT E_T\)	MOVE PROFILE	TRADING IMPLICATION
$\ddot e_t > 0$	Accelerating — slow start, fast finish	Classic liquidation cascade shape. Enter early, scale in. Do not wait for chart confirmation — the geometry shows it before the chart does.
$\ddot e_t = 0$	Constant velocity — linear move	Steady trend. Fixed position size. Linear time stop.
$\ddot e_t < 0$	Decelerating — fast start, slow finish	Impulse-and-fade shape. Enter immediately at full size. Take profit early — the move front-loads its magnitude.

28.9 — Reversal Prediction

THREE POST-THRESHOLD REGIMES — DETERMINED BY R_T BEFORE ARRIVAL

$$ R_t \gg 0 \;\Rightarrow\; \Omega_1\text{ after crossing} \;\Rightarrow\; \text{mechanical reversal} $$

$$ M_\text{reversal} = \lambda_t^\text{price}\cdot\sqrt{\Sigma_0^{(1,1)}}\cdot(\sqrt{e^*} - \sqrt{e_0}), \qquad e_0 = Nd $$

$$ R_t \text{ small positive} \;\Rightarrow\; \Omega_2\text{ after crossing} \;\Rightarrow\; \text{deceleration, no sharp reversal} $$

$$ R_t \to 0 $$

$$\;\Rightarrow\; \Omega_3\text{ after crossing} \;\Rightarrow\; \text{cascade, no reversal, shifted equilibrium}$$

R_t gives you the post-threshold regime **before** you get there. The reversal trade setup or the cascade continuation is known at entry.

28.10 — Entry Criteria (Complete)

USE INSTEAD OF SIMULATION

SIX ENTRY CONDITIONS — ALL MUST BE SATISFIED

$$\begin{array}{ll} \text{(1)} & \dot e_t > 0 \quad\text{(energy growing — move has momentum)} \\[5pt] \text{(2)} & \ddot e_t \geq 0 \quad\text{(energy accelerating — not a fading move)} \\[5pt] \text{(3)} & R_t > R_\text{min} \quad\text{(not approaching uncontrollable regime)} \\[5pt] \text{(4)} & \lambda_t^\text{price} > 0.5 \quad\text{(majority of energy in price dimension)} \\[5pt] \text{(5)} & \dot\lambda_t^\text{price} \geq 0 \quad\text{(energy rotating into price, not out)} \\[5pt] \text{(6)} & \psi_t < \pi/4 \quad\text{(abstract risk real in physical space — not a false alarm)} \end{array}$$

Conditions (1)–(3) are from structural inevitability ranking. Conditions (4)–(6) are the additions that fix the projection problem and eliminate false alarms.

28.11 — Position Sizing (Corrected)

CORRECTED POSITION SIZE — THREE GEOMETRIC MULTIPLIERS

$$\text{size}_t = \text{size}_\text{base} \times \underbrace{(1-\gamma_t^*)}_{\text{blind-spot damping}} \times \underbrace{\lambda_t^\text{price}}_{\text{price loading}} \times \underbrace{\mathbf{1}[\dot\lambda_t^\text{price} \geq 0]}_{\text{energy rotation gate}}$$

The third multiplier is binary — zero size when energy is rotating out of price regardless of how elevated e_t is. This eliminates the entire class of trades where the risk model fires but price doesn't move.

28.12 — Exit Triggers

EXIT WHEN ANY OF THE FOLLOWING

$$\begin{array}{ll} \text{(A)} & \tau_\text{adjusted}\text{ bars elapsed} \quad\text{(time stop)} \\[4pt] \text{(B)} & \text{RSS}_t > 0.7\text{ first crossing} \quad\text{(regime critical)} \\[4pt] \text{(C)} & R_t \leq 0 \quad\text{(entering uncontrollable regime)} \\[4pt] \text{(D)} & \dot\lambda_t^\text{price} < 0\text{ and }\lambda_t^\text{price} < 0.4 \quad\text{(energy leaving price)} \\[4pt] \text{(E)} & \text{Dominant channel switches to }\delta_G \quad\text{(novel regime — edge is gone)} \\[4pt] \text{(F)} & \psi_t > \pi/2 \quad\text{(complete misalignment — signal is false)} \end{array}$$

28.13 — Complete Output Per Bar

OUTPUT	FORMULA	TRADING MEANING
λ_t^price	$\sqrt{e_t^\text{price,full}/e_t}$	Fraction of energy in price — magnitude reliability
$\dot\lambda_t^\text{price}$	Bar-over-bar change in λ_t^price	Energy rotation — into price (enter) or out (wait)
$\text{sign}(M_t)$	Sign of return deviation sum	Direction: long or short
$M_t^\text{remaining}$	$\lambda_t^\text{price}\sqrt{\Sigma_0^{(1,1)}}(\sqrt{e^*}-\sqrt{e_t})$	Expected magnitude — order of magnitude, not exact
$M_t^{(0.05,0.95)}$	Chi-squared confidence interval	Statistically calibrated

OUTPUT	FORMULA	TRADING MEANING
		magnitude zone
τ_adjusted	$\tau_\text{quad}/\lambda_t^\text{price}$	Time stop in bars
$\ddot e_t\text{ sign}$	$g_t^\top\nabla^2 E_\text{BS}\,g_t\cdot\|\dot X_t\|^2$	Move shape: accelerating / linear / decelerating
I_t	$(\lambda_t^\text{price})^2\sqrt{\Sigma_0^{(1,1)}}(\sqrt{e^*}-\sqrt{e_t})/\tau_\text{quad}$	Average intensity per bar
R_t	$\sum_k[g_t]_k\langle\partial_X\delta_k,g_t\rangle_F$	Reversal (positive) vs cascade (near zero)
M_reversal	$\lambda_t^\text{price}\sqrt{\Sigma_0^{(1,1)}}(\sqrt{e^*}-\sqrt{e_0})$	Post-threshold reversal magnitude (when $R_t \gg 0$)
$\cos\theta_t^\text{price}$	$P_t^\text{price}/(\|\tilde G_t^\text{price}\|_F\|F_t\|_F)$	Buy-dip filter: negative = dip supported, positive = avoid
ψ_t	$\arccos(R_t/\|\tilde G_t\|_F\|g_t\|)$	False alarm filter: above

OUTPUT	FORMULA	TRADING MEANING
		$\pi/4$ = signal unreliable
RSS_t	Six-signal geometric mean	Regime state: 0 = normal, 1 = collapse imminent
Dominant a_k	$\max_k(a_k)$	Which channel driving instability: G = novel regime, A = cascade in progress

28.14 — Worked Example: BTC 5-Minute Chart

SETUP

$$N=8,\quad d=8,\quad e^* = \chi^2_{64,0.99} \approx 94,\quad \sqrt{\Sigma_0^{(1,1)}} = 0.02\;(2\%\text{ per bar})$$

CURRENT BAR STATE

$$e_t = 45,\quad e_t^\text{price,full} = 30,\quad \dot e_t = +3.2/\text{bar},\quad \ddot e_t = +0.8/\text{bar}^2$$
$$\lambda_t^\text{price} = \sqrt{30/45} = 0.816,\quad \dot\lambda_t^\text{price} = +0.03 > 0\;(\text{energy rotating into price})$$
$$\text{Return deviation sum} = -0.08\;(\text{downward move}),\quad R_t = +2.1\;(\text{reversal regime})$$

PREDICTIONS

$$ M_t^\text{remaining} = 0.816 \times 0.02 \times (\sqrt{94}-\sqrt{45}) = 0.816 \times 0.02 \times (9.70-6.71) = 0.0488 $$ **4.9% remaining downward move** (corrected from naive 8.4% — energy is 82% in price not 100%) $$ \tau_\text{quad} = \frac{-3.2+\sqrt{10.24+78.4}}{0.8} = 7.8\text{ bars} $$ $$ \tau_\text{adjusted} = 7.8/0.816 = 9.6\text{ bars} = 48\text{ minutes} $$ $$ I_t = (0.816)^2 \times 0.02 \times (9.70-6.71)/7.8 = 0.666 \times 0.02 \times 3.99/7.8 = 0.0068/\text{bar} = 0.68\%\text{ per 5-min bar} $$ Move profile: $\ddot e_t > 0$ → accelerating. Slow start, fast finish. Scale in over first half of τ_adjusted. $$ M_\text{reversal} = 0.816 \times 0.02 \times (\sqrt{94}-\sqrt{64}) = 0.816 \times 0.02 \times 1.70 = 0.028 = 2.8\%\text{ recovery} $$

USE INSTEAD OF SIMULATION

TRADE PLAN FROM GEOMETRY

Entry: short at current bar (all 6 entry conditions satisfied) Direction: down (return deviation sum negative) Target zone: -3.4% to -6.4% from entry ($0.7\times$ to $1.3\times M_t^\text{remaining}$) Time stop: 48 minutes (τ_adjusted) Scale-in: add to short in bars 4–7 (accelerating profile — back-loads magnitude) After target: if $R_t > 0$ still, flip long for 2.8% reversal Exit early if: $\dot\lambda_t^\text{price} < 0$ (energy leaving price), dominant channel switches to δ_G, or $R_t \to 0$

Key correction: Naive formula predicted 8.4% — corrected formula gives 4.9%. The difference is $1 - \lambda_t^\text{price} = 18\%$ of energy sitting in volatility/dispersion, not in price. Trading the naive target would have missed the exit by 3.5%.

28.15 — The Intraday Decision Tree

PER-BAR DECISION LOGIC — EXECUTED IN ORDER

$$ \begin{array}{ll} \textbf{Step 1} & \text{Compute }X_t\text{ (update 64 features)} \\[4pt] \textbf{Step 2} & \text{Compute }e_t, g_t, \tilde G_t, \lambda_t^\text{price}, \dot\lambda_t^\text{price}, \psi_t, R_t, P_t \\[4pt] \textbf{Step 3} & \text{If }R_t \leq 0\text{: EXIT ALL, no entries until }R_t > 0 \\[4pt] \textbf{Step 4} & \text{If }\psi_t > \pi/4\text{: HOLD, signal may be false alarm} \\[4pt] \textbf{Step 5} & \text{If }\dot\lambda_t^\text{price} < 0\text{ and }\lambda_t^\text{price} < 0.4\text{: NO $$

NEW ENTRIES, energy leaving price} \\[4pt] \textbf{Step 6} & \text{Compute RSS}_t\text{ and }\tau_\text{adjusted}, M_t^\text{remaining}, I_t, \ddot e_t\text{ sign} \\[4pt] \textbf{Step 7} & \text{Check all 6 entry conditions — enter only if all satisfied} \\[4pt] \textbf{Step 8} & \text{Set size} = \text{size}_\text{base}\times(1-\gamma^*)\times\lambda_t^\text{price} \\[4pt] \textbf{Step 9} & \text{Set time stop at }\tau_\text{adjusted}\text{ bars, target zone at }[0.7,1.3]\times M_t^\text{remaining} \\[4pt] \textbf{Step 10} & \text{Monitor }R_t\text{ for reversal vs cascade classification} \end{array} \]

28.16 — What This Gives That Does Not Exist Elsewhere

CAPABILITY	STANDARD TOOLS	THIS OPERATOR
Direction	Momentum, trend-following	Sign of return deviation sum — geometric
Magnitude	ATR, Fibonacci	$\lambda_t^\text{price}\sqrt{\Sigma_0^{(1,1)}}(\sqrt{e^*}-\sqrt{e_t})$ — calibrated
Duration	None	$\tau_\text{adjusted} = \tau_\text{quad}/\lambda_t^\text{price}$ — geometric
Move shape	None	Sign of $\ddot e_t$ — before visible on chart
False alarm filter	None	$\psi_t < \pi/4$ and $\lambda_t^\text{price} > 0.5$
Cascade vs reversal	None	R_t sign — before threshold crossed
Energy rotation	None	$\dot\lambda_t^\text{price}$ — when price will move vs when volatility expands
Reversal magnitude	None	$\lambda_t^\text{price}\sqrt{\Sigma_0^{(1,1)}}(\sqrt{e^*}-\sqrt{e_0})$

CAPABILITY	STANDARD TOOLS	THIS OPERATOR
Position size	Fixed or vol-scaled	$\text{base}\times(1-\gamma^*)\times\lambda_t^\text{price}\times\mathbf{1}[\dot\lambda\geq 0]$ — proved minimal
Confidence interval	None	Chi-squared calibrated magnitude zone

Charts show the integral of price velocity over time. This operator computes the velocity, its derivative, its feature decomposition, and its geometric relationship to the energy landscape — before the chart shows any of it. The move profile is known before it is visible. That is not a claim about prediction. It is a statement about what the derivative contains that the integral does not.

§ XXIX — EMPIRICAL ADDENDUM · APRIL 2026

State-Dependent Optimal Thresholds & the 4-Quadrant Attribution Law

This section records the mathematical structure and empirical discoveries that emerged from deploying the BSDT framework on intraday crypto futures (ETH/BTC/SOL, 1-hour bars, 2021–2026). The central result — that *optimal classification thresholds are functions of the system state, not constants* — generalises to any BSDT instantiation.

29.1 — Per-Channel Collapse Attribution

Given the k=1 PCA projection of the state matrix, the k-th channel signal is $\tilde S_k$. The mean-normalised attribution for channel k at time t is:

COLLAPSE ATTRIBUTION (K=1 PCA, 4 CHANNELS C, G, A, T)

$$a_k(t) = \frac{\tilde S_k(t)^2}{\|\tilde S(t)\|^2}, \qquad \tilde S_k = \frac{S_k}{\mu_k^{(\text{normal})}}, \qquad \sum_k a_k = 1$$

The mean-normalisation $\mu_k^{(\text{normal})}$ is estimated once on the calibration window and

- `a_C` Mahalanobis channel attribution — within-ellipsoid crowding
- `a_G` Gap channel attribution — PCA out-of-subspace displacement
- `a_A` Acceleration (velocity) channel attribution — temporal novelty rate
- `a_T` Topological novelty channel attribution — KDE surface departure

frozen. It converts raw channel magnitudes into the fraction of total anomaly energy carried by each channel. $a_G \approx 0.45$ means 45% of the current deviation from normal lives in the gap (PCA out-of-plane) channel.

29.2 — Rolling Memory Operator

A single attribution reading is noisy. The *rolling maximum over the recent window* extracts the peak excitation within a memory horizon N:

MEMORY OPERATOR (SHIFT BY 1 BAR TO ELIMINATE LOOKAHEAD)

$$G_{\text{mem}}(t) = \max_{s \in [t-N, t-1]} a_G(s), \qquad T_{\text{mem}}(t) = \max_{s \in [t-N, t-1]} a_T(s)$$

Optimal calibration on crypto futures: $N = 8$ bars (8 hours). Values of $N \in \{6,\ldots,10\}$ produce Sharpe within 0.05.

29.3 — The 4-Quadrant Phase Modulator

The attribution pair $(G_{\text{mem}}, T_{\text{mem}})$ partitions the space into four quadrants. An *A-channel firing condition* gates whether any quadrant is active. The quadrant indicators are:

QUADRANT INDICATORS (ALL SHIFT-1 TO PREVENT LOOKAHEAD)

$$\begin{aligned} \mathbf{1}_{\text{fire}} &= \mathbf{1}[a_A(t-1) > \theta_A] \\[6pt] q_{GH,TH}(t) &= \mathbf{1}_{\text{fire}} \cdot \mathbf{1}[G_{\text{mem}}(t) > \theta_G] \cdot \mathbf{1}[T_{\text{mem}}(t) > \theta_T] \\ q_{GH,TL}(t) &= \mathbf{1}_{\text{fire}} \cdot \mathbf{1}[G_{\text{mem}}(t) > \theta_G] \cdot \mathbf{1}[T_{\text{mem}}(t) \le \theta_T] \\ q_{GL,TH}(t) &= \mathbf{1}_{\text{fire}} \cdot \mathbf{1}[G_{\text{mem}}(t) \le \theta_G] \cdot \mathbf{1}[T_{\text{mem}}(t) > \theta_T] \\ q_{GL,TL}(t) &=$$

$$\mathbf{1}_{\text{fire}} \cdot \mathbf{1}[G_{\text{mem}}(t) \le \theta_G] \cdot \mathbf{1}[T_{\text{mem}}(t) \le \theta_T] \end{aligned}$$

The four quadrants are assigned scalar multipliers $\beta_{GH,TH}, \beta_{GH,TL}, \beta_{GL,TH}, \beta_{GL,TL} \in \mathbb{R}$. The phase modulator clips to $[\phi_{\min}, \phi_{\max}]$:

PHASE MODULATOR

$$\phi_t = \operatorname{clip}\!\left(1 + \beta_{GH,TH}\,q_{GH,TH}(t) + \beta_{GH,TL}\,q_{GH,TL}(t) + \beta_{GL,TH}\,q_{GL,TH}(t) + \beta_{GL,TL}\,q_{GL,TL}(t),\;\phi_{\min},\;\phi_{\max}\right)$$

$\theta_A = 0.65$	A-channel firing threshold — gates all quadrant activity
$\theta_G = 0.45$	G-channel memory threshold (static base; adapted in §29.5)
$\theta_T = 0.41$	T-channel memory threshold — cliff-sensitive, keep static
$\beta_{GH,TH} = +5.0$	Both channels high → strong boost ($\phi_{max} - 1$)
$\beta_{GH,TL} = \beta_{GL,TH} = \beta_{GL,TL} = -1.0$	All other quadrants → full kill ($\phi \to 0.05$)
$\phi_{max} = 6.0$	Clip ceiling — confirmed optimal at clip = 6
$\phi_{min} = 0.05$	Floor — prevents zero sizing without position flip

The at-boundary structure $\beta_{GH,TH} = \phi_{\max} - 1$ means the boost fires at exactly the clip ceiling — a nonlinear gate implicit in the clamp. Boosting above $\phi_{\max}$ gains nothing while degrading MaxDD.

29.4 — The Full Position Sizing Law

Let base_t be the portfolio base PnL (sum of independent strategies), γ^*_{t-1} the adaptive damping from §VIII, and $\lambda_{\text{pct}}(t-1)$ the lambda percentile from §XXVIII. The complete position sizing law is:

V54 CHAMPION PNL EXPRESSION — ALL TERMS SHIFT-1 TO PREVENT LOOKAHEAD

$$\text{PnL}_t = \text{base}_t \;\cdot\; \underbrace{\operatorname{clip}(1-\gamma^*_{t-1},\,0,\,1)\cdot\left(0.75 + 0.5\,\lambda_{\text{pct}}(t-1)\right)}_{\text{size}_{sym}(t)} \;\cdot\; \underbrace{\left(1 + c_b\,\mathbf{1}[a_G(t-1) > \theta_{GBT}]\right)}_{\text{G-boost}} \;\cdot\; \phi_t$$

`c_b = 0.65`	Champion boost amplitude — additional 65% size when G-channel active
`θ_{GBT} = 0.45`	G-boost activation threshold (equals θ_G in v54)
`size_{sym}`	Symmetric sizing: damped by adaptive γ^*, enlarged by lambda percentile

29.5 — State-Dependent Threshold: The Core Discovery

Regime analysis of the test window (2023–2026) revealed a persistent asymmetry: high-volatility regimes yield Sharpe +4.30 vs +3.43 in low-volatility regimes under the static threshold. This motivates adapting θ_G inversely with the normalised realised volatility:

This is the first result in this document derived from live empirical optimisation rather than from first principles. It generalises: any scalar threshold in §VI–§IX that exists in a system with a measurable volatility / dispersion regime should be treated as a function, not a parameter.

DYNAMIC G-THRESHOLD — V55 LAW

$$\theta_G(t) = \theta_{G,\text{base}} \cdot \bigl(1 + K_G\,\cdot\,(1 - \hat{rv}_t)\bigr)$$

NORMALISED REALISED VOLATILITY (1-WEEK, SHIFT-1)

$$\hat{rv}_t = \frac{\sigma_{168}(r_t)}{\mathbb{E}_{1000}[\sigma_{168}(r_t)]}\Bigg|_{t-1}, \qquad \sigma_{168}(r_t) = \sqrt{\frac{1}{168}\sum_{s=t-168}^{t-1} r_s^2 - \bar r^2}$$

The sign convention is deliberate: $\hat{rv}_t > 1$ (high vol) lowers θ_G — the system trades more readily when the anomaly energy is elevated. $\hat{rv}_t < 1$ (quiet markets) raises θ_G — requiring stronger G-channel evidence before activating the boost regime.

$K_G = 0.10$	Identified as optimal via walk-forward fitting on 2023–2024, confirmed on frozen holdout 2025–2026
$\theta_{G,base} = 0.45$	Static fallback (K_G = 0 recovers v54)
clip range	$\theta_G(t)$ clipped to [0.20, 0.70] — prevents degenerate thresholds

CRITICAL MANIFOLD GEOMETRY

Generalisation principle: For any BSDT scalar threshold $\theta \in \{\theta_G, \theta_T, \theta_A, \theta_C\}$, the state-dependent form is:

$$\theta(t) = \theta_{\text{base}} \cdot f(\text{regime}_t, K)$$

where f satisfies: (i) $f = 1$ at long-run regime average, (ii) f is monotone in regime signal, (iii) $K = 0$ recovers the static baseline. The function class that minimises overfitting risk relative to improvement size is the linear form $f = 1 + K(1 - \text{regime})$, confirmed superior to quadratic and policy-ensemble forms in adversarial testing.

29.6 — Threshold Sensitivity Classification

Not all thresholds respond equally to adaptation. Empirical Sharpe sensitivity to ±0.01 perturbation classifies each threshold:

THRESHOLD	TYPE	±0.01 SHARPE IMPACT	ADAPTATION VERDICT
$\theta_T = 0.41$	Cliff	−0.15 to −0.19	Keep static. High sensitivity → dynamic adaptation pays most here (v57 target)
$\theta_G = 0.45$	Gradual	−0.06 to −0.09	Dynamic via $\hat{rv}_t$. K_G = 0.10, +0.179 Sharpe confirmed OOS

THRESHOLD	TYPE	±0.01 SHARPE IMPACT	ADAPTATION VERDICT
$\theta_A = 0.65$	Mild	−0.04 to −0.10	Mildly sensitive; not yet addressed
$c_b = 0.65$	Flat	< 0.001	Completely flat — do not adapt, hold constant
CLIP = 6.0	Flat	< 0.008	Confirmed plateau at clip ∈ [5.5, 6.5]

The paradox: the *most sensitive* threshold (θ_T) is also the one where dynamic adaptation is most likely to help — state dependence rescues cliff-parameters by moving the operating point away from the cliff before it is reached.

29.7 — Adversarial Validation Methodology (4-Test Kill Suite)

Standard backtests conflate in-sample structural fitting with genuine edge discovery. The following four-test suite is designed to *falsify* a discovered improvement before trusting it. All four tests were applied to the v55 dynamic threshold before promotion:

#	TEST	WHAT IT KILLS	PASS CRITERION	V55 RESULT
1	Walk-Forward OOS	In-sample structural fitting	K fitted on first half; frozen evaluation on second half beats static baseline	OOS Δ = +0.098 Sharpe **STRONG PASS**
2	Lag Corruption	Timing / lookahead artifacts	Sharpe at +48h extra lag > baseline − 0.10	Sharpe at +48h = +3.986 (> static +3.857) **STRONG PASS**
3	Shuffle Null Distribution (N = 500)	Random alignment of regime signal with PnL	p-value < 0.05 (original in top 5% of null)	$p = 0.024$, $z = +1.89\sigma$ **PASS**
4	Formula Robustness	Fragile coincidence with one specific functional form	All 4 forms beat static; peak Sharpe spread < 0.10	Spread = 0.11 (driven by quadratic scale mismatch) **FAIL**

Score 3/4: "Likely real — direction structural, functional form not fully resolved." The v56 form-identification experiment (14 runs) subsequently confirmed the linear form is optimal, the direction gap over inversion is +0.158 Sharpe, and the average-form policy ensemble does not improve over linear.

> SHUFFLE TEST NULL DISTRIBUTION
>
> $$ H_0:\; \text{Sharpe}(\hat{rv}^\sigma) \;\overset{d}{=}\; \text{Sharpe}(\hat{rv}), \quad \hat{rv}^\sigma = \text{random permutation of }\hat{rv}\text{ within test window} $$
> $$ p = \frac{1}{B}\sum_{b=1}^{B}\mathbf{1}\!\left[\text{Sharpe}^{(b)} \ge \text{Sharpe}_{\text{observed}}\right], \quad B = 500 $$

29.8 — Empirical Progression: v34 → v56

VERSION	SHARPE	Δ	KEY STRUCTURAL CHANGE	CAGR (1× LEVER)	MAXDD
v34	+2.616	—	Base portfolio: 6 daily + 2 intraday strategies	+11.2%	−5.1%
v48	+3.616	+1.000	4-quadrant phase modulator, clip = 5, $\beta_{GH,TH} = 4$	+16.6%	−4.1%
v49	+3.674	+0.058	GH,TL = −0.60; A-fire threshold = 0.65; champion boost = 0.60	+17.1%	−3.8%
v50	+3.717	+0.044	GH,TL = −0.80 joint; A-fire = 0.67	+17.5%	−3.9%
v51	+3.730	+0.012	GH,TL converges to −1.00; all non-GH,TH quadrants = full kill	+17.1%	−3.6%
v52	+3.809	**+0.079**	T_THRESH 0.40 → 0.41; clip 5.0 → 6.0 (at-boundary structure β = clip − 1)	+18.6%	−3.8%
v53	+3.856	+0.047	A-fire = 0.65 at new clip/T_THRESH baseline	+19.0%	−3.7%

VERSION	SHARPE	Δ	KEY STRUCTURAL CHANGE	CAGR (1× LEVER)	MAXDD
v54	+3.857	+0.001	*Plateau confirmed*: all static parameter axes exhausted	+19.1%	−3.7%
v55	**+4.036**	**+0.179**	**Dynamic θ_G: first state-dependent threshold (K_G = 0.10)**	+20.2%	−3.7%
v56	+4.036	±0.000	Form ID: linear confirmed optimal; direction gap +0.158 over inversion	+20.2%	−3.7%
v57	+4.036	±0.000	3-test validation: OOS ✓, Vol stress (rv×1.10) ✗ FAIL, T-coupling ✓. Identifies one-sided boundary condition.	+20.2%	−3.7%
v58	**+3.959**	**−0.077**	**Asymmetric clamp lo=−0.01 + EMA(3): rv×1.10 drop −0.297 → −0.074. All 4 robustness tests PASS. Production lock.**	+19.7%	−3.7%

29.9 — What Generalises Beyond Crypto

The following results hold for any domain where the BSDT framework is instantiated and a measurable volatility/dispersion regime exists:

USE INSTEAD OF SIMULATION

GENERALISATION RESULTS

G1. Parameter plateau ≠ signal exhaustion. When all scalar parameter axes are exhausted (Δ < 0.002 Sharpe per perturbation), the correct response is not more parameter search but *promotion of parameters to state-dependent policies*.

G2. Threshold sensitivity predicts adaptation potential. A threshold with high two-sided sensitivity (\(|\partial \text{Sharpe}/\partial\theta| \gg 0\)) is a cliff — and a cliff at the right operating point is exactly where a state-dependent threshold provides the largest benefit by dynamically avoiding the cliff.

G3. Linear functional form is prior. Until proven otherwise, use \(\theta(t) = \theta_{\text{base}}(1 + K(1 - \text{regime}_t))\). Richer forms (quadratic, tanh, exp) introduce scale mismatches that require additional K optimisation without measurable gain. The direction of adaptation (sign of K) is the structural discovery; the form is secondary.

G4. The at-boundary structure ($\beta = \phi_{\max} - 1$) is the correct GH,TH boost calibration. Setting the boost to exactly reach the clip ceiling maximises signal intensity without regime ambiguity. Values $\beta > \phi_{\max} - 1$ are clipped and waste parameter budget.

G5. Walk-forward OOS + lag corruption are the two non-negotiable tests. Shuffle confirms significance. Formula robustness confirms form stability. A result that passes tests 1 and 2 but fails 3 and 4 may still be real; a result that fails tests 1 or 2 is not deployable regardless of in-sample Sharpe.

29.10 — v58 Stability Patch: Asymmetric Clamp + EMA Smoothing

Problem (v57 Test 2 failure). The v55 formula $\text{adj} = K_G(1 - \hat{r}v_t)$ has no floor. When $\hat{r}v_t$ is persistently inflated by +10% (rv×1.10 stress), adj becomes increasingly negative, $\theta_G(t)$ drops below its optimal value, and the system overtrades. Sharpe drop: −0.297 (FAIL, threshold 0.15).

Asymmetry of the boundary condition. The failure is one-sided: rv×0.90 produces only −0.113 (PASS); additive noise (±N(0,0.02)) produces −0.122 (PASS). Only sustained upward rv bias triggers the failure. This identifies a directional boundary, not fragility — the upside path (lower threshold in high-vol) carries structural edge but an unbounded floor.

USE INSTEAD OF SIMULATION

V58 PATCH EQUATIONS

$$\hat{r}v_t = \text{EWM}(\hat{r}v_t^{\text{raw}},\,\tau=3) \mid_{t-1}$$

$$\text{adj}_t = \text{clip}\!\left(K_G(1 - \hat{r}v_t),\; l_0,\; +0.10\right)$$

$$\theta_G(t) = \text{clip}\!\left(\theta_{\text{base}}\,(1 + \text{adj}_t),\; 0.20,\; 0.70\right)$$

Production values: $K_G = 0.10$ (frozen), $\tau = 3$, $l_0 = -0.01$, $\theta_{\text{base}} = 0.45$.

Semantics of \(l_0 = -0.01\): floor on adj → \(\theta_G(t) \geq 0.45 \times 0.99 = 0.4455\) regardless of rv stress. Upper bound \(+0.10\) unchanged (low-vol rise is safe).

Clamp sweep (EMA τ=3 fixed)

\(L_0\)	FULL SHARPE	OOS H2 (25-26)	RV×1.10 DROP	VERDICT
−0.05	+4.016	+4.800	−0.180	FAIL
−0.04	+4.014	+4.800	−0.166	FAIL
−0.03	+4.003	+4.800	−0.166	FAIL
−0.02	+4.035	+4.861	−0.185	FAIL (non-monotonic)
−0.01	**+3.959**	**+4.740**	**−0.074**	**PASS ← champion**
0.00	+3.917	+4.690	−0.105	PASS

Note: non-monotonic behaviour at \(l_0 = -0.02\) (worse than \(-0.03\)) is a local data interaction. The PASS threshold jumps discontinuously at \(l_0 = -0.01\).

v58 validation results (all 4 robustness tests)

CONFIG	FULL SHARPE	OOS H2	RV×1.10 DROP	OOS ✓	STRESS ✓
v55_ref (no clamp, no EMA)	+4.036	+4.800	−0.297	✓	✗
v58_clamp (lo=−0.05, no EMA)	+4.036	+4.800	−0.222	✓	✗
v58_ema (lo=−0.05, EMA=3)	+4.016	+4.800	−0.180	✓	✗
v58_tight (lo=−0.01, EMA=3)	**+3.959**	**+4.740**	**−0.074**	✓	✓

USE INSTEAD OF SIMULATION

V58 PRODUCTION SPECIFICATION (LOCKED)

rv_norm: $\hat{r}v_t = \text{EWM}(\sigma_{1h,\text{168h}} / \bar{\sigma}_{1000h},\;\tau=3)\mid_{t-1}$

adj: $\text{clip}(0.10 \times (1 - \hat{r}v_t),\;-0.01,\;+0.10)$

θ_G(t): $\text{clip}(0.45\times(1+\text{adj}),\;0.20,\;0.70)$

Frozen parameters: CLIP=6.0, GH_TH=5.0, GH_TL=GL_TH=GL_TL=−1.0, N=8, T_THRESH=0.41, CB=0.65, AFT=0.65, K_G=0.10

Full-period Sharpe: +3.959 | OOS 2025–2026 Sharpe: +4.740 | MaxDD: −3.7% | rv×1.10 drop: −0.074

Extracted from `energy.py` (GravityEngine), `bsdt.py` (Blind-Spot Detection Theory) & `network.py` (Ledoit-Wolf Correlation Network) · Unified field: Quadsurf $S^\top AS$ + Expogate $\sum e^{\beta_k S_k}$ + Signed LR $w^\top S$ · Sections §I–XXIX · Crypto application §XXVIII · State-dependent thresholds §XXIX · Corrected definitions §XXIV · Per-channel Lyapunov §XXV · ψ_t §XXVI · Domain discoveries §XXVII · defaults: α=0.10, γ=1.00, σ=1.00, λ=0.10, $\varepsilon=10^{-5}$, θ=1.00, F_max=100 · CCyB [0,250] bps · v58 champion: clip=6, θ_G(t)=clip(0.45(1+clip(0.10(1−r̂v_t),−0.01,+0.10)),0.20,0.70), τ_EMA=3, Sharpe=+3.959, OOS-H2=+4.740